BOTANY DOWNS
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AS 91890 – Conduct an Inquiry to propose a Digital Technology Outcome

Student Name: Krish Ram
Student NSN #: 0144176710

Declaration:

I Krish Ram declare that this assessment is my own work, except where acknowledged appropriately (e.g., use of referencing). I declare that I composed the writing and/or translations in this assessment independently, using the tools and resources defined for use in this assessment. I am aware that any breach of this statement or identified academic misconduct will be followed up and may result in disciplinary action.

Signed

Krish Ram

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Introduction:

Critical inquiry cycle:

What is a critical inquiry cycle?

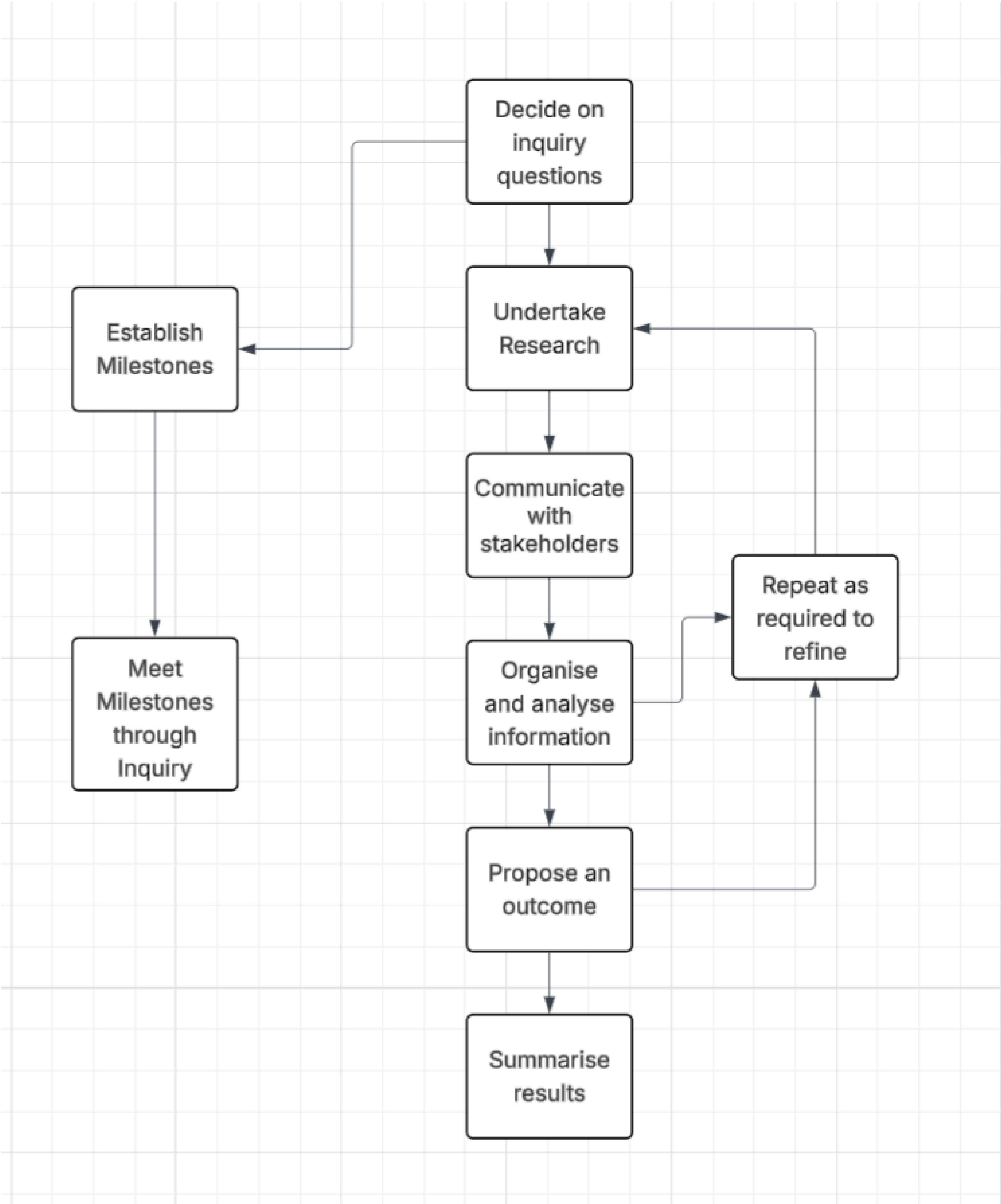
Critical Inquiry is a method of exploring and understanding problems through systematic investigation opposed to enquiry where you would ask less in-depth questions. It starts when we observe something interesting or identify problems that needs solving. I will then narrow down the list and arrive at the most important and burning problems that need immediate solutions which could make a big difference to the community. Instead of accepting things at face value, we must ask questions, investigate deeper, and test our ideas to gain new insights.

What makes Critical Inquiry powerful is that it's driven by our own curiosity and genuine desire to understand or solve problems. When we're personally invested in finding answers, we're more likely to dig deeper and think more creatively.

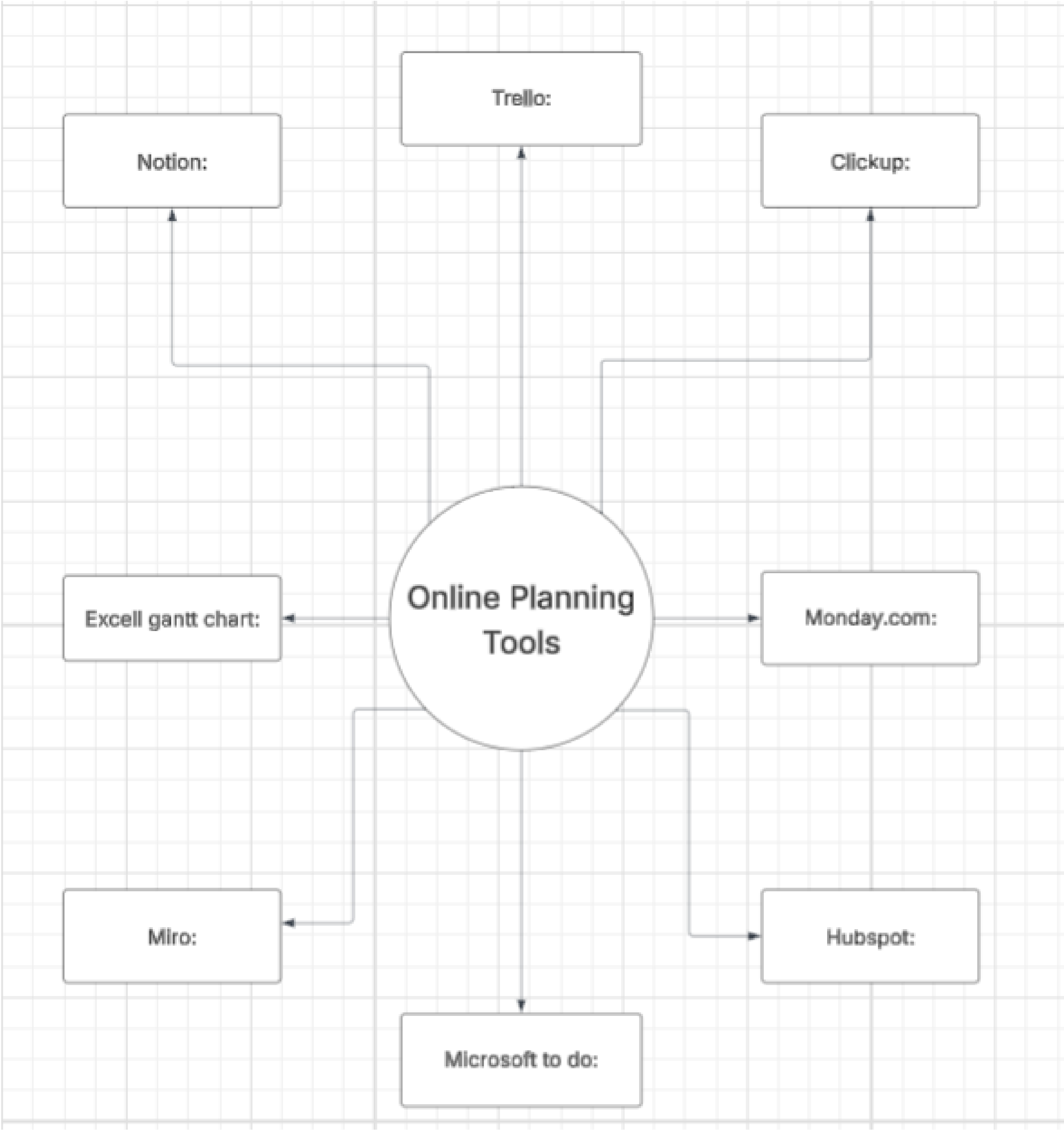
To strengthen my investigation, I will conduct thorough research into existing solutions to understand what's already available and how effectively these solutions are addressing the problem. I will actively involve key stakeholders who will be directly impacted by my proposed solution, as their input and feedback will be crucial to developing an effective outcome. Throughout this process, I will utilize various planning tools to organize my investigation, ensuring it goes beyond simply asking questions and fact-checking.

This structured approach will help me develop a comprehensive understanding of the problem and guide me toward creating the most effective solution. For this project I'll be following the process outlined in the image below, which provides a structured framework for my investigation.

Image of Critical Inquiry Cycle:



Research of planning tools:



3 positives and negatives for each of the Online Project Planning tools:

Here's a breakdown of each tool's key pros (+) and cons (-):

Notion

+ Highly customizable and flexible

+ Excellent documentation capabilities

+ Great for both personal and team use

- Steep learning curve

- Can become messy without proper organization

- Sometimes overwhelming with too many features

Trello

+ Simple, intuitive interface

+ Great visual organization with cards

+ Easy to get started

- Limited in complex project management

- Basic free version

- Can become cluttered with many tasks

Clickup

+ Comprehensive feature set

+ Multiple view options

+ Good customization options

- Complex interface

- Can be slow at times

- Too many features for simple projects

Excel Gantt Chart

+ Familiar interface for most users

+ Complete control over formatting

+ No additional cost if you have Excel

- Manual updates required

- Limited collaboration features

- Time-consuming to maintain

Monday.com

+ Visually appealing interface

+ Strong collaboration features

+ Multiple project views

- Expensive for small teams

- Can be overwhelming initially

- Limited free version

Miro

+ Excellent for visual planning

+ Great for brainstorming

+ Strong collaboration features

- More suited for visual work than task management

- Can be expensive

- Limited organizational features

Microsoft To Do

+ Clean, simple interface

+ Good integration with Microsoft products

+ Free with Microsoft account

- Limited project management features

- Basic functionality only

- Not ideal for complex projects

Hubspot

+ Good CRM integration

+ Strong marketing features

+ Comprehensive analytics

- Primarily focused on sales/marketing

- Expensive for full features

- Overkill for simple project management

My chosen Online Project Planning Tool:

I have chosen to use Trello as my online planning tool as I am experienced with this application. Trello's intuitive card-based interface makes it easy to organize my project tasks and track progress visually. Its simple drag-and-drop functionality allows me to quickly adjust priorities and move tasks between different stages of completion. The ability to add checklists, due dates, and labels to cards will help me stay organized throughout my project development. While Trello may have limitations for complex project management, its features are well-suited for my current needs and will help me maintain a clear overview of my project timeline and deliverables.

Trello Link:

https://trello.com/invite/b/697fd80b584b2481647a7e32/ATTI7a48c9d888ad37c71a3b6428657531b5321137AB/inquiry-project

List of tasks:

1. Setting Up Milestones
2. Defining inquiry
3. Choosing a project management tool
4. Advantages and disadvantages of PMT
5. Set up Trello
6. Writing list of tasks

Milestone 1:

7. Brainstorming the issues
8. Choosing 3 issues
9. Choosing an inquiry topic
10. Open ended Inquiry Question

Milestone 2:

11. Writing inquiry questions

12. Research 3 articles on the Inquiry topic

12. Analyse & Record Information

13. Refine the Inquiry Question

Milestone 3:

14. Research on Refined question 3 articles

15. Analyse & record information(Table will be given)

16. Final Refinement of Inquiry Question.

Milestone 4:

17. Research Existing solutions

18. Write about the relevant inplications

19. Proposal

20. Stakeholder Profiling

21. Preparing questionnaire for stakeholders (on proposing app and their requirements)

Milestone 5:

22. Risks and Risk Mitigation

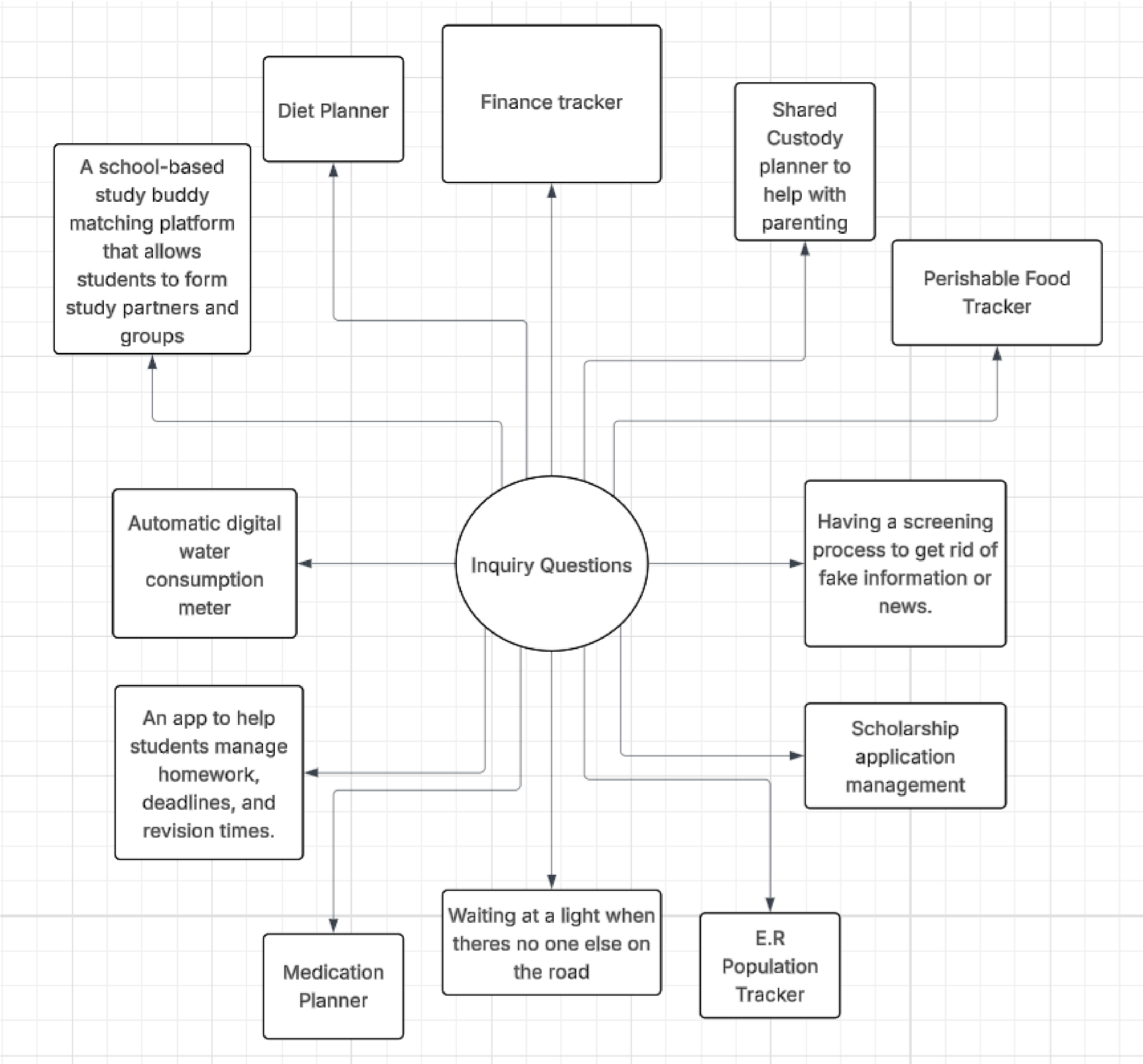
23. Evaluation of the Proposed outcome(Strengths, weakness, Further opportunities)

24. Write a summary of the inquiry process

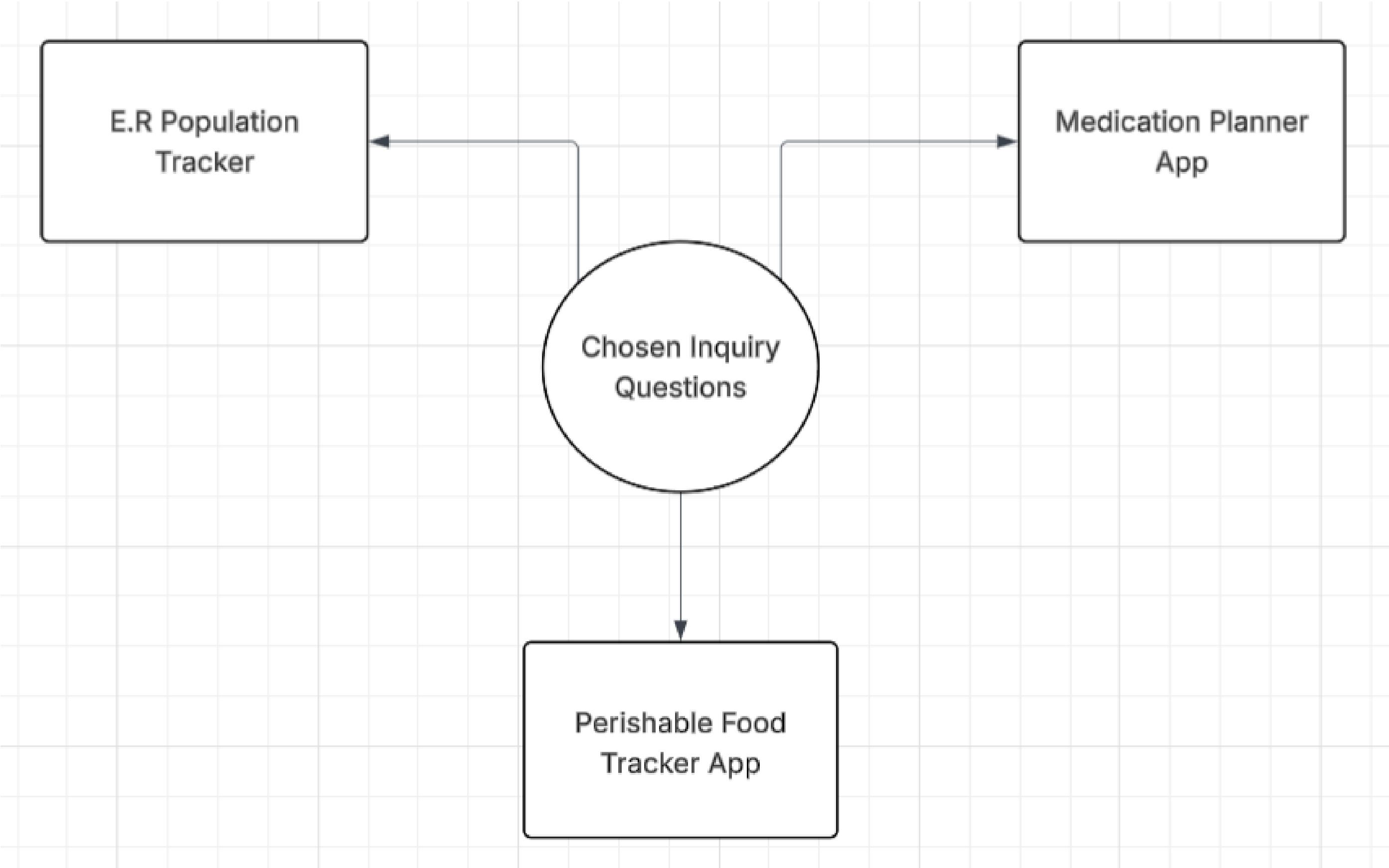
25. Acknowledgements

Milestone 6:

My Brainstorm of Different Issues:



My Narrowed Down Brainstorm:



The gaps within the 3 inquiry questions:

When brainstorming potential topics for my inquiry, I identified three distinct issues: a Perishable Food Tracker App, a Medication Planner App, and an E.R Population Tracker App. While each presented an opportunity to explore a digital solution, I needed to evaluate which one would lead to the richest and most meaningful inquiry.

- The Perishable Food Tracker App addresses the problem of food waste by helping users track expiry dates. However, the gap here is primarily personal, and the solution relies entirely on user-generated data. From an inquiry perspective, this limits the scope for deeper investigation. The questions it raises are mostly about user interface design and behavior change, with little need to explore external factors, conflicting viewpoints, or ethical dilemmas. It is a practical idea, but the inquiry would be narrow.
- Similarly, the Medication Planner App aims to help users manage complex medication schedules through reminders. Again, the data is user-generated, and while privacy is a consideration, the inquiry would largely focus on features and usability rather than wider systemic issues.
- The E.R Population Tracker addresses a significant public gap. Currently, when people face an urgent medical issue, they have no way of knowing which Emergency Department has the shortest wait time. This leads to overcrowding, patient frustration, and delays in care. Unlike the other ideas, this solution requires sourcing live data from external authorities such as

District Health Boards (Te Whatu Ora). This immediately opens up a rich field of inquiry questions: Is this data publicly accessible? How reliable and accurate is it? What are the privacy implications of broadcasting live patient loads? Furthermore, the stakeholders are diverse and hold conflicting perspectives. Patients want transparency to make informed choices, while hospital managers may worry about data being misinterpreted. Ambulance services, triage nurses, and policymakers would all have unique and valuable viewpoints to consider.

In conclusion, while the Perishable Food Tracker and Medication Planner are worthwhile topics, they lead to a narrower set of questions focused on individual users. The E.R Population Tracker, however, opens up a much wider inquiry into data ethics, stakeholder perspectives, and public health infrastructure. For these reasons, it is the most compelling focus for my investigation and would benefit the most amount of people, not just one niche.

My Chosen Inquiry Topic:

E.R Population Tracker

My Big Question (Theme/Problem/Issue):

The theme I have chosen is public access to real-time healthcare information, specifically Emergency Department (ER) wait times and patient loads. From this theme, I will brainstorm the reasons why this information is not readily available, the consequences of this information gap for the public, and how a digital solution could address this. The end result of my inquiry will be a proposal for a digital outcome.

My Inquiry Statement:

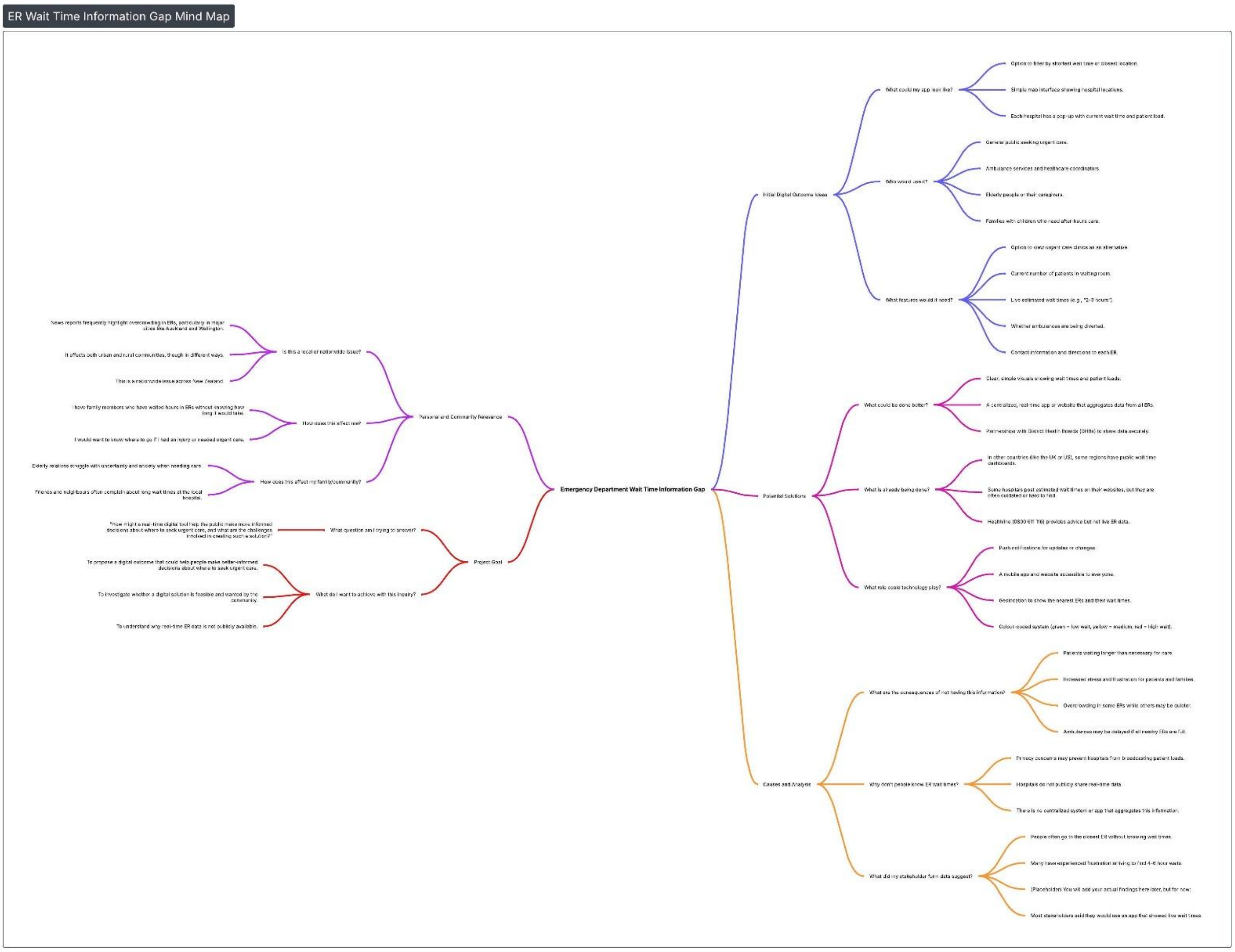
"New Zealanders are struggling to make informed decisions about where to seek urgent care due to a lack of accessible, real-time data on Emergency Department wait times and patient populations, leading to overcrowding in some ERs and delays in treatment for all."

The pressure on New Zealand's public health system, particularly its Emergency Departments, is a growing concern. When individuals have a medical issue that is urgent but not a life-threatening emergency (e.g., a deep cut, a high fever, a broken bone), they often have no choice but to go to their local hospital's ER and wait, sometimes for many hours. Currently, there is no centralized, public-facing system that shows which ERs are busiest, which have the shortest estimated wait times, or the current patient load. This lack of information means people cannot make choices that could get them faster care or help balance the load across different health providers, including urgent care clinics. This "information blackout" contributes to patient frustration, potentially worsens health outcomes due to delayed care, and puts unnecessary strain on already overwhelmed emergency departments

Initial Brainstorm:

Link:

https://lucid.app/lucidchart/37c12062-4a8e-429e-99e7-196a9881996d/edit?view_items=jokCRy604kXS&page=0_0&invitationId=inv_f685c956-19c0-4ac0-ade6-cd179d8aa2f9



I first inquired on the topic question of "What aspects of the theme are relevant to me, and my friends, family and/or community?" To answer this I thought about my own experiences visiting the Emergency Department with family members and the frustration of not knowing how long we would wait. I also thought about conversations I have heard from friends and neighbours who complain about long waits at our local hospital. This helped me realise that this is not just a personal issue but something that affects many people in my community. I also considered that this is a nationwide issue across New Zealand, with news reports frequently showing overcrowding in ERs, particularly in major cities like Auckland and Wellington. This shows that the problem is significant and widespread.

I then went further in-depth into this inquiry thinking for the brainstorm and asked myself "What causes it? Why does it happen? How does it happen?" This was the main topic of this brainstorm. I thought of all the reasons why people cannot access real-time ER wait times and why this information gap exists. To answer this I considered that hospitals do not publicly share real-time data, there is no centralized system or app that aggregates this information, and there may be privacy concerns preventing hospitals from broadcasting patient loads. I also reflected on what my stakeholder form data suggested about the consequences of this gap. From my initial form responses, I found that people often go to the closest ER without knowing wait times, many have experienced frustration arriving to find 4-6 hour waits, and most stakeholders said they would use an app that showed live wait times. This confirmed that the problem is real and that there is demand for a solution.

This then leads to the next inquiry I brought up which was "What can be done about it? What is being/has been done? What could I do?" Due to my main cause being a lack of accessible real-time data, I started thinking about what solutions already exist. I considered that some hospitals post estimated wait times on their websites, but they are often outdated or hard to find. In other countries like the UK or US, some regions have public wait time dashboards, but nothing like that exists in New Zealand. Healthline provides advice but not live ER data. From this I realised that what could be done better is a centralized, real-time app or website that aggregates data from all ERs through partnerships with District Health Boards. Technology could play a key role through a mobile app and website with geolocation to show the nearest ERs and their wait times, using a simple colour-coded system.

This then intertwined into bringing up "How can I link this to a digital outcome? What are my initial ideas about what I might propose?" I looked at digital outcomes that could address the needs identified in my stakeholder form. My initial ideas would be a simple map interface showing hospital locations with pop-ups displaying current wait time and patient load. The app would need features such as live estimated wait times, current number of patients in the waiting room, whether ambulances are being diverted, and contact information for each ER. I also considered including urgent care clinics as an alternative option. The users would be the general public seeking urgent care, families with children who need after-hours care, elderly people or their caregivers, and potentially even ambulance services and healthcare coordinators.

To finish off my brainstorming I then asked myself "What is the goal of my project?" I then answered this being that the goal of this project is to understand why real-time ER data is not publicly available, to investigate whether a digital solution is feasible and wanted by the community, and to propose a digital outcome that could help people make better-informed decisions about where to seek urgent care. The core question I am trying to answer is: "How might a real-time digital tool help the public make more informed decisions about where to seek urgent care, and what are the challenges involved in creating such a solution?"

Initial Inquiry Question:

From my initial brainstorming, stakeholder meetings, and my inquiry statement I will be making the inquiry question of: **"Why do New Zealanders struggle to access real-time information about Emergency Department wait times, and how could a public digital tool help them make more informed decisions about where to seek urgent care?"**

Initial Stakeholder Meetings

To gather initial opinions on my inquiry question and proposed digital outcome, I created a questionnaire using Microsoft Forms and distributed it to a range of stakeholders — including students, teachers, and other community members. I aimed to get a variety of ages and backgrounds to ensure my findings were not limited to one group. The form was designed to answer two core questions: (1) do people actually struggle with not knowing ED wait times, and (2) would they use a real-time public tool if one existed? I collected 9 responses in total. Although stakeholders were not interviewed in person, the form acted as a structured meeting, allowing me to capture their perspectives systematically and compare them across different demographic groups.

Form Questions and Results Summary:

I asked the following questions in my form. The results are summarised below and analysed in relation to my inquiry question:

Q1 — What is your age range? 6 respondents were Under 18, 2 were aged 25–34, and 1 was 65 and older. This age spread is skewed younger due to the school-based distribution of the form, but having an older respondent (65+) and two working-age adults (25–34) adds useful perspective from groups who are more likely to visit EDs. I am aware that a wider age range would strengthen the findings, and this is acknowledged as a limitation of my stakeholder sample.

Q2 — Have you or an immediate family member visited an ED in the past 5 years? 6 out of 9 respondents (67%) said yes. This confirms that the majority of my stakeholders have direct personal experience with the issue I am investigating, which increases the relevance and credibility of their responses. The 3 who said no still provided useful perspective on how a prospective non-user might perceive the tool.

Q3 — How did you decide which ED to go to? 7 out of 9 said they went to the closest one. Only 2 said they already had a preferred hospital. Nobody reported using any digital tool, checking online for wait times, or comparing hospitals before going. This is a critical finding: it directly confirms the problem I identified in my initial brainstorm — that patients are currently making their ED choice purely based on proximity, with no information about wait times or capacity. This strongly supports the case for my proposed outcome.

Q4 — Have you ever arrived at an ED only to find the wait was much longer than expected? 6 said “Yes, occasionally,” 2 said “Yes, frequently,” and 1 was “Unsure.” Nobody answered “No, never.” This means 100% of respondents have experienced, or cannot rule out, arriving to find an unexpectedly long wait. This is a striking result that directly validates the core problem my inquiry is investigating.

Q5 — How frustrating is not knowing ED wait times? (1–5 scale) Responses: 3× rated 4/5, 3× rated 5/5 (extremely frustrating), 3× rated 3/5. The average rating was 4.0 out of 5. No respondent rated this below 3. The three 5/5 responses came from stakeholders with the most direct ED experience (Hayden Chung, Antony Hamer, and Teodor Iordan — all of whom reported frequent or occasional unexpected long waits). This high frustration score reinforces that the problem is not merely inconvenient but is a genuine pain point for people who have experienced it.

Q6 — How likely would you be to use a free public app showing live ED wait times? 2 said “Very likely,” 5 said “Likely,” 1 said “Unsure,” and 1 said “Unsure” (Teodor Iordan — the same respondent who said he would not trust the data). Nobody responded “Unlikely” or “Very unlikely.” 7 out of 9 respondents (78%) said they would be likely or very likely to use the app. This is a strong validation of demand for my proposed outcome. The 2 uncertain responses are also valuable: they highlight that data trustworthiness is a key design consideration, which is directly addressed in my proposal’s requirements section.

Q7 — What information would be most valuable in such an app? All 9 respondents selected “Estimated wait time” — making it the unanimous top priority. “Current number of patients in the waiting room” was selected by 7, “Number of available beds” by 5, and “Whether ambulances are being diverted elsewhere” by 2. One respondent (Antony Hamer) added “prices” as a custom response, which is interesting and reflects an awareness that private urgent care clinics charge fees — a consideration for a future iteration of the app. These results directly shaped the core display requirements of my proposed outcome: estimated wait time and waiting room patient count as the primary data points, with bed availability as a secondary feature.

Q8 — Would you trust the accuracy of live data provided directly from hospitals? 7 out of 9 said “Yes, but I would still want to confirm by calling,” 1 said “Unsure,” and 1 (Teodor Iordan) said “No, I would not trust it.” This result is nuanced and important: while there is general willingness to use the data, most stakeholders want a verification option rather than relying on the app alone. This supports my proposal’s design decision to always include a hospital phone number and the 111 call button alongside wait time data, so users are never fully dependent on the live data alone.

Overall Findings from Initial Stakeholder Meetings:

The stakeholder responses strongly confirm all three core assumptions behind my inquiry. First, that the problem is real: 100% of respondents have experienced or cannot rule out arriving at an unexpectedly busy ED, and the average frustration rating was 4.0/5. Second, that current behaviour is uninformed: 7 of 9 people simply go to their nearest ED with no comparison of alternatives. Third, that there is genuine demand for my proposed solution: 78% said they would be likely or very likely to use a free live wait time app. The one respondent (Teodor Iordan) who would not trust the data and was unsure about using the app provides an important counterpoint — it reminds me that data trustworthiness and transparency must be built into the product from the start, not treated as an afterthought. The unanimous selection of “estimated wait time” as the most valuable data point confirms this should be the primary feature of the WaitSmart NZ interface.

Stakeholder Profiles:

The following table summarises the profile and key responses of each of my 9 stakeholders, following the format used in Cuda Wong's exemplar inquiry.

Name	Age Range	ED Experience	Unexpected Long Wait?	Frustration (1-5)	Would Use App?	Trusts Hospital Data?	Top Info Wanted
Sheroy Bhathena	Under 18	No	Yes, occasionally	4 / 5	Very likely	Yes (would still call to confirm)	Wait time; Available beds
Linda Byrne	25–34	Yes	Yes, occasionally	3 / 5	Likely	Yes (would still call to confirm)	Wait time; Patient count
Ryan Yang	Under 18	No	Unsure	3 / 5	Likely	Yes (would still call to confirm)	Wait time; Available beds
Benjamin Lau	Under 18	Yes	Yes, occasionally	4 / 5	Likely	Yes (would still call to confirm)	Wait time; Ambulance diversions; Patient count; Beds
Edrian Lee	Under 18	No	Yes, occasionally	4 / 5	Likely	Yes (would still call to confirm)	Wait time; Patient count; Beds
Hayden Chung	Under 18	Yes	Yes, frequently	5 / 5	Very likely	Yes (would still call to confirm)	Wait time; Patient count
Antony Hamer	25–34	Yes	Yes, frequently	5 / 5	Likely	Unsure	Wait time; Patient count; Prices

Teodor Iordan	Under 18	Yes	Yes, occasionally	5 / 5	Unsure	No — would not trust it	Wait time; Patient count; Ambulance diversions
Alan Taylor	65 and older	No	Yes, occasionally	3 / 5	Unsure	Unsure	Wait time only

What the Stakeholder Profiles Tell Me:

Looking across the profiles, several clear patterns emerge. The two stakeholders with the highest frustration scores (5/5) — Hayden Chung and Antony Hamer — are also the ones who have experienced unexpected long waits most frequently. This pattern confirms that frequent negative ED experiences directly correlate with higher demand for a wait time tool. Alan Taylor, the oldest respondent (65 and older), was the most cautious overall: unsure about using the app and unsure about trusting the data, and only interested in the single piece of information — estimated wait time. This is valuable because it tells me that the 65+ age group, who are among the most likely to need emergency care, may require an especially simple, single-purpose interface. Teodor Iordan's profile stands out as the most sceptical: he rated frustration 5/5 (so acknowledges the problem is real) but would not trust the data and was unsure about using the app. This suggests that for some users, the barrier is not relevance but trust — reinforcing the proposal's commitment to transparency, data timestamps, and a clear explanation of data sources. Benjamin Lau was the most engaged respondent, selecting all four data types including ambulance diversions, which suggests that more technically aware younger users may want a richer feature set. A future version of the app could offer a "detailed view" mode for users like Benjamin alongside the simple colour-coded default view for users like Alan.

Here is a link to my form:

3DIP ER Population Tracker – Collaboration

Research - Introduction & research process & Organise research information:

To identify how to create a solution to my initial inquiry question I have done research on existing systems for tracking emergency department wait times, the challenges hospitals face in sharing real-time data, and the potential benefits of a public-facing digital tool. Also, on why New Zealanders currently struggle to access this information and how other countries or regions have approached similar problems. In addition, I have done some research on possible methods to create this solution and existing apps that are already trying to address this gap. I have analysed all of these sources for useful information as well as their reliability.

The reason why I am doing research is to see if people/companies from the wider community have had a similar problem/situation in which the public cannot access real-time information about emergency department wait times and patient loads, leading to frustration, overcrowding, and delays in care. This is a growing issue in New Zealand's health system so I feel the research for this will be readily available from news articles, government reports, and existing digital solutions. The process of research I will be using, is a common process used partially in Science and English reports, the steps/questions I will be following is as such:

Research process steps:

1. Understand the question (define the task)

What do I need to do? What is my purpose? Identify the keywords

Understand exactly what the question is asking, so I know how to answer it really well.

2. Locate sources of information

Find great resources quickly and easily with some simple search techniques.

What do I already know? What do I need to found out? What sort of information do I need?

3. Select what information I use

What information do I/don't I need? How relevant is the information? Is the information from a reliable source?

Learn how to choose the best, most reliable information to use in my assignment.

The following resources will help me decide what information is relevant to my question and if the information is credible:

Evaluating Websites and other information:

Skim and scan reading:

Read the first paragraph or introduction of the resource in full.

Read the topic sentence, this is usually the first sentence of the paragraph.

Read the conclusion in full. This is usually the last paragraph. Decide whether what I have found is useful enough to read in full.

Ask myself, will I be able to answer my questions by reading this resource?

4. Organise my findings

Do I have all the information I need? Have I collected enough information to answer my question? Have I referenced all my information?

Turn my information into a good set of notes that will make the writing process quicker and easier. Before I begin to write my notes:

Make sure I have understood what I have read, Make sure what I have read is relevant Think about the most important points and how they fit together

Once I begin writing my notes:

Keep my notes brief (bullet- pointed lists are best) Do not copy out large pieces of text Concentrate of the key points in the text Keep a note of the source I am using

Research 1:

Author	RNZ
Title:	Wellington Hospital Emergency Department went into code red nearly twice a day in 2025
Name of Publication:	NZ Herald
Link:	https://www.nzherald.co.nz/nz/wellington-hospital-emergency-department-went-into-code-red-nearly-twice-a-day-in-2025/KD6A72JUCJGUPFVDU32YP5EPKI/#:~:text=An%20Official%20Information%20Act%20response,hours%2C%20below%20the%20Government's%20target.
Date Published	21 Jan, 2026 04:54 PM
Describe the Article Briefly	This article reports on data released under the Official Information Act revealing that Wellington Hospital's Emergency Department declared its most critical "code red" status an average of nearly twice a day between January and October 2025, totalling 575 code reds. It shows that only about half of patients were treated and discharged or admitted within six hours, falling below the Government's target. The article also notes that over 3,200 patients left without being treated during this period. It includes perspectives from a Wellington Hospital emergency nurse, Labour's Health spokesperson Dr Ayesha Verrall, and Health Minister Simeon Brown, discussing the causes (staffing shortages, infrastructure) and the Government's planned responses (funding for redevelopment and increased staffing).
What are one or two key points that this source can contribute to my research?	The article provides powerful, official data proving that New Zealand emergency departments are under immense strain, with "code red" events occurring frequently. This directly supports the premise of my inquiry that there is a critical need for tools to help manage patient flow and inform the public. The statistic that over 3,200 people left without being treated (about 10 people a day) is particularly significant. It strongly suggests that if these individuals had access to real-time information about wait times or patient loads at other departments, they might have sought care elsewhere, potentially receiving treatment and reducing pressure on Wellington Hospital.
What is one direct quote I can use as evidence?	"Wellington Hospital's emergency department went into its most critical code-red status on average nearly twice a day between January and October last year... More than 3200 patients left the emergency department without being treated - around 10 people a day."
Is this a reliable	The NZ Herald is one of New Zealand's largest and most established news outlets, with editorial standards for fact-checking and accuracy. The core data in the article

source of information ? How do I know? (Accuracy, reliability, bias)	comes from an Official Information Act (OIA) response, which is a formal, legal process for releasing government information. This makes the specific statistics (575 code reds, 3200 patients leaving) highly reliable and verifiable. The article also includes multiple perspectives: from the hospital management (Jamie Duncan's statement), a frontline nurse (anonymous), the opposition (Dr Ayesha Verrall), and the government (Minister Simeon Brown). This inclusion of different viewpoints reduces the risk of bias, as it presents a more complete picture of the situation rather than a single agenda
Which part of my inquiry does this information support? (relevance/significance)	This information is highly relevant to establishing the significance and urgency of the problem I am investigating. It provides concrete evidence of the consequences of the information gap my inquiry focuses on. The fact that patients are leaving without being seen directly connects to the core question of my inquiry: could a real-time digital tool have helped these patients make a different, more informed decision? It supports the need for a solution that empowers the public with information.
Have any of my thoughts or ideas changed after reading or viewing this text?	Reading this article has reinforced the severity of the ED overcrowding issue but also showed the complexity of the problem. While I initially focused on the patient's lack of information, the article shows that underlying causes also include staffing shortages and infrastructure limits. This means my proposed digital solution is not a standalone fix but one part of a larger system. It would not solve the root causes of the code reds, but it could be a vital tool to help mitigate the effects by helping to distribute patient load more evenly and reduce the number of people who leave without care.
Interesting/implications for consideration-how will this information inform my digital outcome?	This information has several key implications for my proposed digital outcome. First, it confirms a clear user need: the 10 patients a day who leave without being seen are a primary audience who could benefit from real-time information. Second, the perspectives from staff and officials suggest the solution needs to be credible and integrated. The app's data would need to be trusted by both the public and health officials. The mention of government funding for EDs and after-hours care suggests that partnering with or aligning with Health NZ's initiatives could be a pathway for the app's development and data access. Finally, the nurse's comment about patients in corridors shows that "wait time" is complex; my outcome might need to define what its data represents (e.g., time to triage, time to see a doctor, current patient load in waiting room).

Research 2:

Author	Valentin Poirot, Antoine Tran, Emma Boudier, Emma Fauvet, Herve Haas, Hanna Vrzakova
Title:	Clinical data integration and processing challenges in healthcare caused by contemporary software design
Name of Publication:	DIGITAL HEALTH (Sage Journals) / PMC (NIH)
Link:	https://pmc.ncbi.nlm.nih.gov/articles/PMC12461051/
Date Published	24th September 2025
Describe the Article Briefly	This peer-reviewed academic article details the process and significant challenges of integrating ten years of clinical data from multiple legacy software systems at the Lenval Children's University Hospital in Nice, France, to create a Clinical Data Warehouse (CDW) for research. The hospital's Emergency Department used four separate, unconnected software platforms (for patient administration, clinical notes, laboratory results, and imaging). The researchers detail the extensive data cleaning required—handling typos, inconsistent formats, missing values, and unwritten clinical "rules"—to make the data usable for analysis and machine learning. They argue that the original design of healthcare software, which prioritises immediate clinical use over future data sharing and analysis, is the root cause of these integration difficulties. The study successfully integrated data from over 575,000 ED admissions
What are one or two key points that this source can contribute to my research?	This article is critically important as it provides expert, evidence-based insight into the primary technical barrier to my proposed outcome: accessing and integrating live data from hospital systems. It confirms that hospitals typically use multiple, incompatible software systems that are not designed to share data externally or even internally. The "four separate software platforms" used for one ED is a clear example. The article also shows the immense complexity of cleaning and standardising "messy" clinical data—a task the authors describe as "complex, labor intensive, and time consuming, and hence, out of the reach of hospital clinicians." This directly informs the feasibility of my proposal, shifting my focus from "if we can build an app" to "how we could potentially access and standardise the underlying data."
What is one direct quote I can use as evidence?	"Healthcare systems such as Terminal Urgences® have been developed and regulated as medical device… Consequently, these systems are rigid, and data collection is not designed to be statistically readable. These systems produce heterogeneous data that are error prone and unsuitable for secondary use."
Is this a reliable source of information? How do I know? (Accuracy, reliability, bias)	This source is highly reliable for an academic inquiry. It is a peer-reviewed article published in a reputable journal (DIGITAL HEALTH) and archived by the U.S. National Institutes of Health (PMC). The authors are a mix of data scientists and clinicians who conducted the research firsthand at a major hospital, giving them direct, practical expertise. The article is thorough, transparent about its methods, and discusses both its successes and the limitations and challenges encountered. It is not an opinion piece but a detailed account of a real-world project, making its findings and conclusions about healthcare data challenges very credible and applicable to my inquiry.
Which part of my inquiry does this	This information directly supports the investigation into the feasibility and challenges of my proposed digital outcome. While my stakeholder data

information support? (relevance/significance)	may show public demand for an ER tracker, this article reveals the significant, non-obvious technical hurdles on the supply side. It supports the part of my inquiry that must consider data sourcing, integration, and standardisation. It validates that the core idea of my proposal is not simple and that any realistic solution would require deep collaboration with health authorities and a sophisticated understanding of their data infrastructure.
Have any of my thoughts or ideas changed after reading or viewing this text?	Yes, my understanding of the problem has deepened significantly. I initially thought the main challenge would be getting permission to access data. This article shows that even if permission were granted, the data itself is likely stored in multiple, incompatible, "legacy" systems and in a "messy" format that is not readily usable. The "labour intensive" process of cleaning data described by the researchers was a reality check. It made me realise that my proposed digital outcome is not just an app development project, but a complex data integration project. This means the skills and resources needed are greater than I first assumed.
Interesting/implications for consideration-how will this information inform my digital outcome?	This information has several major implications for my proposed outcome. Firstly, it suggests that a phased approach might be necessary. A full, live, multi-hospital tracker is a very long-term goal. An initial outcome could focus on one department or one hospital to prove the concept and work through the data integration challenges. Secondly, it shows the absolute necessity of partnering with Health New Zealand (Te Whatu Ora) from the very beginning. My proposal cannot be a solo project; it would need to be a collaborative effort with the data owners and the technical teams who understand these legacy systems. Thirdly, the article's focus on "secondary use" of data is key. My proposal would need to clearly define its purpose (e.g., public information, not clinical decision-making) and address the privacy and security concerns that are central to discussions about using patient data for any secondary purpose. Finally, the article's insights into unwritten clinical practices (like dropping the trailing zero in blood pressure) show that any data display would need to be carefully designed so that it is not misleading to the public.

Research 3:

Author	CDA-AMC (Canada's Drug Agency and Health Technology Agency)
Title:	Technologies to Address Wait Times in the Emergency Department
Name of Publication:	Canadian Journal of Health Technologies
Link:	https://canjhealthtechnol.ca/index.php/cjht/article/view/EN0058#:~:text=Digital%20information%20tools%20provide%20real,by%20ordering%20tests%20or%20imaging.
Date Published	July 17, 2025
Describe the Article Briefly	This is a formal health technology review published by a Canadian government agency. It provides an overview of emerging technologies specifically designed to reduce wait times and manage patient safety in Emergency Departments. The report categorises and analyses four key technology types: AI triage and clinical decision support models, portable/wearable vital sign monitors, digital information tools (which provide real-time updates on wait times, patient volumes, and available resources to both patients and staff), and telemedicine. It discusses the potential impacts of these technologies, such as improved patient flow, better resource utilization, and enhanced safety for waiting patients. Crucially, it also outlines the significant evidence gaps and implementation challenges that remain before widespread adoption, including the need for robust studies, regulatory frameworks, and addressing data privacy and algorithm biases.
What are one or two key points that this source can contribute to my research?	This source is directly relevant as it validates the core concept of my proposed outcome. The report explicitly lists "digital information tools" that provide real-time updates on wait times and patient volumes as a key technology category for addressing ED strain. It confirms that my proposed solution is not just a good idea, but is being actively considered by health systems internationally as a legitimate tool for improvement. The article also provides a balanced perspective by detailing the "What Else Do We Need to Know?" section, which shows the current lack of large-scale studies, the need for regulatory frameworks, and concerns about data privacy. This gives me critical information to discuss the challenges and future possibilities for my proposed outcome.
What is one direct quote I can use as evidence?	"Digital information tools provide real-time updates to patients or clinicians working in the ED about average wait times, patient volumes, and available resources… AI triage systems and digital information tools could improve patient flow by streamlining triage or redirecting patients with low-acuity care needs to choose alternative ED sites."
Is this a reliable source of information? How do I know? (Accuracy, reliability, bias)	This source is exceptionally reliable for my inquiry. It is published by the Canadian Journal of Health Technologies, which is a peer-reviewed, academic publication. The authors are CDA-AMC (Canada's Drug Agency and Health Technology Agency) , an official Canadian government agency responsible for health technology assessment. Reports from such agencies are produced through rigorous, evidence-based processes to inform government policy and clinical practice. The article is a formal technology review, not an opinion piece, and it presents a balanced view by including both the potential benefits and the current limitations and research gaps. This government and academic backing makes it a highly authoritative source.
Which part of my inquiry does this information	This information supports multiple parts of my inquiry. Most directly, it supports the justification for my proposed digital outcome by showing that health experts and government agencies recognise digital information tools as a valid solution to ED overcrowding. It supports the "Future Possibilities" and "Areas for Improvement" sections by outlining the next steps needed for these technologies

support? (relevance/ significance)	(robust studies, regulations). It also supports the "Risks" and "Ethical Implications" sections by showing the key concerns of data privacy and the potential for bias in AI-powered systems, which are relevant even if my initial proposal doesn't use complex AI.
Have any of my thoughts or ideas changed after reading or viewing this text?	Yes, this article has broadened my perspective. I initially focused on the patient-facing app idea. This report shows that the technology landscape is wider, with tools for staff (like dashboards) and even AI triage systems. It reinforces that my idea is part of a global conversation. More importantly, the article's emphasis on the lack of robust real-world studies and the need for regulatory frameworks has made me realise that proposing a digital outcome is not just about the tech, but also about the evidence and governance needed to make it trustworthy and implementable. This has shifted my thinking from "building an app" to "proposing a solution that fits within a complex, evidence-based health system."
Interesting/implications for consideration-how will this information inform my digital outcome?	This information has several key implications. Firstly, it provides a powerful piece of external validation: my proposed outcome aligns with what health technology agencies are already investigating. I can use this to argue for its relevance. Secondly, the article's finding that these tools can help "redirect patients with low-acuity care needs to choose alternative ED sites" directly supports the primary benefit I want to achieve. Thirdly, the "What Else Do We Need to Know?" section provides a ready-made list of considerations for my own proposal. In discussing the feasibility, I must acknowledge the need for more research, clear data governance, and privacy protections. It suggests that my proposal should not be presented as a finished product, but as a well-researched concept that, with proper partnership and testing (a pilot study), could address a critical health system need. The note that technology works best "in tandem with human experience" is also a crucial reminder that my digital tool should be designed to support, not replace, clinical staff and patient decision-making.

My Initial Inquiry question: **"Why do New Zealanders struggle to access real-time information about Emergency Department wait times, and how could a public digital tool help them make more informed decisions about where to seek urgent care?"**

Has now been refined to:

Refined Inquiry Question:

"What are the key technical, ethical, and regulatory challenges involved in developing a real-time digital tool to display Emergency Department wait times in New Zealand, and how might these challenges be addressed to create a feasible and trustworthy public information system?"

Why This is a Stronger, Refined Question:

Research Source:	Key Insight:	How It Refined My Question:
Research #1 (NZ Herald)	The problem is real and severe (575 code reds, 3200 patients left untreated).	My question must still address the core problem (why a tool is needed), but it now also needs to consider the urgency and scale of the issue.
Research #2 (PMC Article)	The technical barriers are massive. Hospital data is stored in multiple, incompatible legacy systems and is "messy," requiring complex cleaning and integration.	My question must now include an investigation into technical feasibility. It is not just about "if" we can build an app, but "how" we can access and standardise the underlying data from complex hospital systems.
Research #3 (Canadian Tech Review)	Digital information tools are recognised internationally as a valid solution, but there are significant evidence gaps and need for regulatory frameworks, data privacy, and addressing bias.	My question must now incorporate the ethical and regulatory dimensions. A trustworthy tool needs more than just good code; it needs governance, privacy protections, and a clear understanding of its limitations.

Further analysis and comparison of perspectives:

I have analysed 3 different articles on the challenges of accessing real-time Emergency Department wait times, the technical barriers to sharing this data, and the potential of digital information tools to address ED overcrowding.

Articles one and three both confirm that long ED wait times are a severe and urgent problem, and that digital information tools are a recognised solution. Article one, a New Zealand news article based on official OIA data, provides local evidence of the crisis—575 code reds and 3200 patients leaving without being treated at Wellington Hospital alone. This establishes the real-world need for my proposed outcome within my own community. Article three, a Canadian government health technology review, validates that digital information tools providing real-time wait times and patient volumes are being actively investigated by health systems internationally as a legitimate strategy to improve patient flow and redirect low-acuity patients. Both articles agree on the potential of my proposed solution, but article three adds a crucial layer by discussing the evidence gaps, regulatory frameworks, and data privacy concerns that must be addressed before such tools can be implemented. This helps my inquiry move beyond simply identifying a problem to understanding the complexities of a real-world solution.

Article two takes a different but complementary angle. While articles one and three focus on the problem and the solution from a public and policy perspective, article two dives deep into the technical "behind the scenes" reality. It is a peer-reviewed academic study detailing the immense difficulty of integrating and cleaning data from the multiple, incompatible legacy software systems used in hospitals. This article directly informs the feasibility part of my refined inquiry question: "What are the key technical challenges?" It reveals that even if hospitals wanted to share data, the data itself is stored in rigid, error-prone systems not designed for "secondary use" like a public app. This contrasts with the relatively straightforward assumption in articles one and three that data could be made available. Article two's findings are critical because they show the single biggest practical barrier to my proposed outcome and suggest that any solution would require deep collaboration with hospital IT departments and data standardisation efforts.

Articles three and two share a focus on the challenges of implementation, though at different levels. Article three discusses the need for regulatory frameworks, policies for accountability, and addressing algorithm biases and data privacy—the governance and ethical challenges. Article two discusses the raw technical challenges of data formats, missing values, and incompatible systems. Together, they paint a complete picture of the hurdles: the data is technically "messy" (article two), and even if cleaned, it needs to be handled within a robust ethical and regulatory framework to be trusted (article three). Article one, by contrast, focuses on the human impact of the problem, which justifies why overcoming these technical and regulatory hurdles is worthwhile.

Articles one, two, and three are what I should consider the most when developing my proposed digital outcome. Article one grounds my proposal in a genuine local need with powerful statistics I can use to argue for its relevance. Article two provides a realistic understanding of the technical complexity, which will inform my thinking about the resources, partnerships (e.g., with Health New Zealand), and phased approach (e.g., starting with a single pilot site) that would be necessary. Article three offers a blueprint of the considerations for making the tool trustworthy and effective, including the need for clear data governance, privacy protections, and an understanding that technology works best alongside human care. Together, these articles will help me produce a proposal that is not just a good idea, but one that is critically informed by the technical, ethical, and systemic realities of the healthcare environment.

Research 4:

Author	Reza Rabiei and Sohrab Almasi
Title:	Requirements and challenges of hospital dashboards: a systematic literature review
Name of Publication:	BMC Medical Informatics and Decision Making
Link:	https://link.springer.com/article/10.1186/s12911-022-02037-8
Date Published	8th November 2022
Describe the Article Briefly	This is a literature review, meaning it rigorously analysed 54 separate research articles published between 2000 and 2020. The study's goal was to identify the functional and non-functional requirements, as well as the implementation challenges, of using dashboards in hospitals. It categorises dashboard requirements into functional (what the dashboard should do, like reporting, alerts, tracking, customisation) and non-functional (how well it should perform, like speed, security, ease of use, integration). Crucially, it also categorises the challenges into four areas: data sources and generation, dashboard content, dashboard design, and implementation and integration. The review found that the majority of dashboard studies were conducted in Emergency Departments, showing the specific relevance of this tool to my inquiry topic.
What are one or two key points that this source can contribute to my research?	This source provides a comprehensive, evidence-based "blueprint" for designing a dashboard like my proposed ER Population Tracker. It tells me exactly what features are essential (functional requirements) and what performance and security considerations are critical (non-functional requirements). The "tracking" feature is directly relevant to monitoring patient flow and identifying crowded wards. The "alert creation" feature could be used to notify users of extremely high wait times. More importantly, the detailed breakdown of challenges and proposed solutions gives me a ready-made framework for discussing the risks and mitigations for my own proposal. For example, the challenge of integrating data from various source systems directly reflects the technical barrier identified in my Research #2 article.
What is one direct quote I can use as evidence?	"Dashboards, by providing information in an appropriate manner, can lead to the proper use of information by users. In order for a dashboard to be effective in clinical and managerial processes, particular attention must be paid to its capabilities, and the challenges of its implementation need to be addressed."
Is this a reliable source of information? How do I know? (Accuracy, reliability, bias)	This source is highly reliable. It is published in BMC Medical Informatics and Decision Making, a reputable, peer-reviewed academic journal. The study is a systematic literature review, which is a robust research methodology that follows a strict protocol (PRISMA) to identify, evaluate, and synthesise all available evidence on a topic. This minimises the bias of relying on a single study. The authors searched four major scientific databases and assessed the quality of the included studies. The fact that 54 articles were synthesised to reach these conclusions makes the findings far more robust and generalisable than a single opinion piece or case study.

Which part of my inquiry does this information support? (relevance/significance)	This information directly supports the core development phase of my inquiry: proposing a digital outcome. It provides a detailed, research-backed list of requirements that my proposed ER Population Tracker should meet. It also directly supports the "Risks and Mitigation" section of my proposal by providing a comprehensive, pre-categorised list of challenges (data sources, content, design, implementation) and even offers research-based solutions for each. This allows me to move from simply identifying risks to proposing well-informed mitigation strategies. The finding that most dashboard studies are in the ED context confirms the strong relevance of this research to my specific topic.
Have any of my thoughts or ideas changed after reading or viewing this text?	Yes, significantly. This article has transformed my understanding of my proposed outcome from a simple "app idea" to a sophisticated "information system" with specific, research-backed requirements. I now understand that a successful dashboard is not just about displaying data, but about carefully considering features like customisation (allowing users to see what's relevant to them), alerts (for critical situations), and tracking (to monitor flow). The systematic categorisation of challenges has also shown me that the barriers are not just technical (data sources) but also relate to design (user interface, visualisation), content (choosing the right indicators), and implementation (integration, user training). This has given me a much more structured and professional framework for my own proposal.
Interesting/implications for consideration-how will this information inform my digital outcome?	This article provides a research-backed blueprint for my proposed ER Population Tracker. The functional requirements it identifies, such as tracking, alerts, and customisation, directly inform the core features my app would need. For example, tracking would allow users to see patient loads across different hospitals, while alerts could notify them of critically long wait times. The non-functional requirements, including speed, security, and ease of use, are equally important considerations for ensuring my proposed outcome is trustworthy and accessible, particularly for stressed or elderly users. The article's categorisation of challenges into data sources, content, design, and implementation gives me a structured framework for my risks and mitigation section. To address data integration challenges, I can draw on the suggested solutions such as creating a data warehouse and using web service architecture. The emphasis on stakeholder participation in selecting appropriate indicators reinforces the need to collaborate with Health New Zealand and clinicians. Finally, the discussion on visualisation best practices, including colour-coding systems, will guide how I propose to display wait times in a way that is instantly understandable and not misleading.

Research 5:

Author	Lorenzo Porta, Franco Aprà, Andrea Caccioppo, Felice Catania, Cinzia Colombo, David Fluck, Giulia Irene Ghilardi, Katarina Hricova, Isaac John, Zora Lazúrová, Jonah Lynch, Giovanni Nattino, George Notas, Chiara Pandolfini, Giovanni Porta, Vicky Rubini, Pankaj Sharma, Lok Thapa, Guido Bertolini
Title:	Enhancing electronic health records with real-time dashboards to improve patient care and service efficiency: a protocol
Name of Publication:	Frontiers in Disaster and Emergency Medicine
Link:	https://www.frontiersin.org/journals/disaster-and-emergency-medicine/articles/10.3389/femer.2025.1558499/full
Date Published	17th December 2025
Describe the Article Briefly	This article describes the protocol for a major European research project called eCREAM, involving 32 Emergency Departments across six countries. The project aims to develop and test three different real-time dashboards fed by electronic health record data: one for ED staff (clinicians and nurses), one for healthcare policymakers and administrators, and crucially, one for citizens. The citizen dashboard is designed to provide the public with real-time information on ED crowding levels and waiting times, with the explicit goal of empowering patients to make better decisions and ultimately reduce overcrowding. The study outlines the technical methods for data integration and standardisation across diverse hospital systems, the process for designing the dashboards with expert panels, and the plan for assessing their usefulness through questionnaires and interviews.
What are one or two key points that this source can contribute to my research?	This source is arguably the most directly relevant to my inquiry. It provides evidence that a large-scale, international, and funded research project is currently underway to create the exact type of digital outcome I am proposing. The eCREAM project validates that my idea is not just a hypothetical student project, but a recognised solution being actively developed by healthcare professionals and researchers across Europe. The article explicitly states that a citizen dashboard could "reduce the self-presentation of patients with minor or non-urgent conditions by empowering them through access to relevant information" and could even help "out-of-hospital emergency organizations divert ambulances to the most suitable ED" based on real-time wait times. This directly supports the core benefits I have identified for my own proposed outcome. Furthermore, the article's detailed methodology for data standardisation, integration, and dashboard design provides a real-world, expert-informed template I can reference in my own proposal.
What is one direct quote I can use as evidence?	"Regarding input factors, dashboards could reduce the self-presentation of patients with minor or non-urgent conditions by empowering them through access to relevant information. A real-time dashboard could also help out-of-hospital emergency organizations divert ambulances to the most suitable ED, not simply in terms of the hospital's clinical expertise concerning the patient's conditions (as is usually the case), but also in terms of the average waiting and treatment times at the different EDs."
Is this a reliable source of information? How do	This source is highly reliable. It is published in Frontiers in Disaster and Emergency Medicine, a reputable, peer-reviewed academic journal. The article is a formal study protocol for a project funded by the European

I know? (Accuracy, reliability, bias)	Commission, UKRI, and SERI, which indicates it has undergone rigorous scrutiny to secure international research funding. The authors are a large, multi-disciplinary team from universities and hospitals across Europe, bringing collective expertise in emergency medicine, epidemiology, and health informatics. The protocol is transparent about its methods, its limitations (e.g., using retrospective data for initial assessment), and its risks. This level of detail, combined with the scale of the project (32 EDs across six countries), makes it a highly credible and authoritative source.
Which part of my inquiry does this information support? (relevance/significance)	This information supports nearly every part of my inquiry. It directly validates the core concept of my proposed digital outcome (a public-facing ED wait time dashboard). It supports my justification by showing that this idea is considered important enough to warrant a multi-million dollar international research project. It supports the "Future Possibilities" section by demonstrating that my proposal aligns with cutting-edge work being done in Europe. It also supports the technical feasibility discussion by outlining real-world methods for data integration and standardisation, reinforcing the points made in my Research #2 and #4 articles. Finally, the planned assessment of the citizen dashboard by real users provides a model for how the success of my own proposed outcome could eventually be measured.
Have any of my thoughts or ideas changed after reading or viewing this text?	Yes, this article has significantly shifted my perspective. Seeing a large, funded European project pursuing the same goal as my inquiry has been incredibly validating. It confirms that my proposed outcome is not only feasible but is considered a priority by healthcare systems abroad. It has also broadened my understanding of the potential impact. I had primarily focused on helping individual patients. The eCREAM project shows that such a tool could also have system-level benefits, such as helping ambulance services make data-driven decisions about where to take patients and allowing policymakers to monitor regional epidemiological trends in real-time. This has given me additional, powerful arguments for the value of my proposed outcome. The article's approach of developing three separate dashboards (for staff, policymakers, and citizens) also made me realise that a comprehensive solution might require more than just a public app, a point I can acknowledge in my proposal's "future improvements" section.
Interesting/implications for consideration-how will this information inform my digital outcome?	This article provides powerful validation and practical guidance. The fact that a major European project is developing a citizen dashboard to reduce ED crowding confirms that my proposed outcome addresses a real, internationally recognised problem. The article's specific goal of empowering patients to choose alternative services and helping to divert ambulances gives me concrete, expert-backed benefits to include in my proposal. The detailed methodology for data standardisation, including data mapping and adopting common formats, reinforces the technical complexity identified in earlier research but also shows that there are established pathways to address it, such as using a central data warehouse and standardised data dictionaries. The planned use of expert panels and user assessment (questionnaires and interviews) provides a model for how I could argue for a human-centred design process and future evaluation of my own outcome. Finally, the project's funding by the European Commission demonstrates that this type of digital health solution is seen as a worthwhile investment, which strengthens the argument for the long-term value and feasibility of my proposal.

Research 6:

Author	Hao Wang, Nethra Sambamoorthi, Devin Sandlin, Usha Sambamoorthi
Title:	Interpretable machine learning models for prolonged Emergency Department wait time prediction
Name of Publication:	BMC Health Services Research
Link:	https://link.springer.com/article/10.1186/s12913-025-12535-w
Date Published	18th March 2025
Describe the Article Briefly	This is a research study that used machine learning (ML) to predict which patients are likely to experience prolonged wait times (over 30 minutes) in the Emergency Department. The researchers analysed data from 177,665 patients over a three-year period and tested five different ML algorithms. The study found that nearly half of all patients (48.20%) experienced prolonged wait times. The key features that predicted longer waits were ED crowding status and the patient's mode of arrival (e.g., ambulance vs. private vehicle). Importantly, the study went beyond just measuring prediction accuracy; it used advanced techniques like SHAP values and partial dependency plots to explain why the models made certain predictions, ensuring the results are interpretable for clinicians.
What are one or two key points that this source can contribute to my research?	This source is invaluable for several reasons. First, it provides strong, data-driven confirmation that ED crowding is the top predictor of prolonged wait times. This directly validates the central problem my proposed dashboard aims to address. Second, it introduces the concept of using machine learning not just to display current wait times, but to predict future wait times based on incoming patient data and current crowding levels. This suggests a more advanced version of my proposed outcome could include predictive features. Third, the study's finding that nearly half of all patients wait over 30 minutes supports the urgent need for a solution like mine. Finally, the study's rigorous approach to model interpretation (using SHAP values) provides a model for how any predictive tool must be transparent and explainable to be trusted by users.
What is one direct quote I can use as evidence?	"The top features influencing patient wait times and gaining the top ranked interactions were ED crowding condition and patient mode of arrival."
Is this a reliable source of information? How do I know? (Accuracy, reliability, bias)	This source is highly reliable. It is published in BMC Health Services Research, a reputable, peer-reviewed academic journal. The study is based on a large sample size (177,665 patients) over a three-year period, which gives the findings statistical power and reduces the chance of anomalous results. The authors used robust ML methodologies (five different algorithms, cross-validation, hyperparameter tuning) and transparently reported their performance metrics. The study also acknowledges its limitations (single-centered, retrospective, potential missing data) which is a sign of good research practice. The authors declare no competing interests, further supporting the objectivity of the findings.
Which part of my inquiry does this information support? (relevance/significance)	This information supports multiple parts of my inquiry. It directly supports the "Key Findings" section by providing an evidence-based list of the most significant factors influencing ED wait times (crowding, mode of arrival, patient demographics). This allows me to ground my proposed dashboard's features in empirical research rather than just speculation. It also supports the "Refined Inquiry Question" by showing that predicting

	wait times (and therefore helping people make decisions) is a complex but feasible problem that advanced analytics can address. The study's emphasis on model interpretability also supports the "Ethics" and "Usability" sections, as it highlights that any AI-driven tool must be explainable to be trusted by clinicians and the public.
Have any of my thoughts or ideas changed after reading or viewing this text?	Yes, this article has significantly deepened my understanding of what a wait time dashboard could become. I initially thought of a simple display of current wait times. This study shows that a more powerful tool could use machine learning to predict future wait times based on factors like current crowding levels, time of day, and incoming patient characteristics. This could be far more useful for a patient deciding where to go. It also made me realise that displaying a single "wait time" is an oversimplification. The study's focus on ESI-3 patients (medium acuity) shows that wait times vary significantly by patient type. A truly useful dashboard would need to either show wait times by triage category or make clear that the displayed time is for a specific level of urgency. The emphasis on interpretability (SHAP values) also taught me that for any AI-driven feature, the system must be able to explain why it made a certain prediction to be trusted.
Interesting/implications for consideration-how will this information inform my digital outcome?	This article has several key implications. The finding that ED crowding is the top predictor of wait times reinforces the core value proposition of my dashboard: helping patients avoid overcrowded EDs. The study's use of a 30-minute threshold for "prolonged wait" gives me a concrete, research-backed benchmark to reference in my proposal. The finding that mode of arrival (ambulance vs. private vehicle) is a major factor suggests my dashboard could provide different information or filters based on how the user is travelling. The successful application of ML models to predict wait times suggests that my proposed outcome could evolve beyond a simple display tool into a predictive decision-support tool. However, the study's emphasis on model interpretability (SHAP, PDP) is a crucial caution: if my proposal ever included predictive AI, it must be transparent and explainable. Finally, the study's limitations (single-centered, retrospective) remind me that any such tool would need to be validated in the New Zealand context and ideally piloted in one region before wider rollout. The authors' note that "individualized wait time predictions are essential for improving ED operational management efficiency" perfectly captures the ultimate goal of my proposed digital outcome.

Overall Research Analysis:

Emergency Departments across multiple countries, including New Zealand, are facing severe overcrowding and prolonged wait times. Research #1 (NZ Herald) provided local evidence that Wellington Hospital experienced 575 code red events in 2025 and over 3,200 patients left without being treated, confirming this is a critical issue within my own community. Research #6 (Wang et al.) supported this by finding that nearly half of all patients (48.20%) in their study experienced wait times exceeding 30 minutes, demonstrating that prolonged waits are a widespread, measurable problem.

There is a significant gap in public access to real-time ED data, and digital information tools are recognised as a viable solution. Research #3 (CDA-AMC) explicitly listed "digital information tools that provide real-time updates to patients about average wait times, patient volumes, and available resources" as a key technology category for addressing ED crowding. Research #5 (Porta et al.) validated this by describing a major European project (eCREAM) actively developing a citizen dashboard to "reduce the self-presentation of patients with minor or non-urgent conditions by empowering them through access to relevant information."

The technical challenges of creating such a tool are substantial, primarily due to the nature of hospital data systems. Research #2 (Poirot et al.) provided detailed evidence that hospital data is stored in multiple, incompatible legacy software systems that are "rigid, and data collection is not designed to be statistically readable." Research #4 (Rabiei and Almasi) confirmed this, categorising "data sources and data generation" as a primary challenge, noting that data is often "heterogeneous, error prone, and unsuitable for secondary use." Research #5 (Porta et al.) offered a practical solution, describing the use of data standardisation, data mapping, and a central data warehouse to overcome these integration barriers.

Beyond simply displaying data, effective dashboards require careful attention to design, functionality, and user needs. Research #4 (Rabiei and Almasi) identified key functional requirements including tracking, alerts, customisation, and reporting, as well as non-functional requirements such as speed, security, and ease of use. Research #3 (CDA-AMC) emphasised that "technological advancements, no matter how promising, cannot replace human care and intuition," and that proposed models work best "in tandem with human experience and empathy." Research #5 (Porta et al.) demonstrated the importance of involving end-users (clinicians, policymakers, and citizens) in the design process through expert panels and focus groups to ensure the dashboard meets actual needs.

Machine learning and predictive analytics offer significant potential to enhance ED wait time tools beyond simple real-time displays. Research #6 (Wang et al.) successfully demonstrated that machine learning models can predict which patients are likely to experience prolonged wait times, with ED crowding status and patient mode of arrival being the top influencing features. The study also emphasised the importance of model interpretability using SHAP values and partial dependency plots, ensuring predictions can be understood and trusted by clinicians. Research #3 (CDA-AMC) noted that "AI triage systems and digital information tools could improve patient flow by streamlining triage or redirecting patients with low-acuity care needs to choose alternative ED sites."

Ethical, privacy, and regulatory considerations are critical for any successful implementation. Research #2 (Poirot et al.) highlighted the sensitivity of clinical data and the need for strict anonymisation, noting that even metadata such as "sex, age, and postal code combined are sufficient to identify a patient's identity in 87% of cases." Research #3 (CDA-AMC) called for "regulatory frameworks, guidelines, and policies to address accountability, errors, algorithm biases, and data privacy concerns." Research #5 (Porta et al.) demonstrated that these concerns can be addressed through careful study design, including data anonymisation at the source, secure transmission protocols, and adherence to regulations like GDPR (General Data Protection Regulation law).

Overall, these articles collectively highlight the urgent need for a real-time ED information tool, the technical and ethical complexities involved in creating one, and the proven pathways (data standardisation, user-centred design, predictive analytics) that can guide its development. They confirm that my proposed digital outcome is not only a valid idea but aligns with international efforts to use technology to improve emergency care, reduce overcrowding, and empower patients to make informed decisions.

Style of Sources & Specific differences:

The sources I have collected vary in style and format, ranging from peer-reviewed academic journal articles to news reports and government health technology reviews. Each source serves a distinct purpose in my inquiry and contributes different types of evidence to support my proposed digital outcome.

Research #1 (NZ Herald) is a mainstream news article based on official data released under the Official Information Act. Its style is journalistic, written for a general audience, and it provides powerful, locally relevant statistics about the severity of ED overcrowding in New Zealand. This source grounds my inquiry in a real, current problem within my own community, making the need for a solution tangible and urgent.

Research #2 (Poirot et al., PMC) is a peer-reviewed academic article detailing the technical process of integrating clinical data from legacy hospital systems. Its style is highly technical and methodical, written for an audience of data scientists, health informaticians, and researchers. This source provides deep, expert insight into the "behind the scenes" challenges of accessing and standardising hospital data, which is critical for understanding the feasibility of my proposed outcome.

Research #3 (CDA-AMC, Canadian Journal of Health Technologies) is a formal government health technology review. Its style is authoritative, concise, and evidence-based, summarising the current state of knowledge on technologies to address ED wait times. As an official government document, it carries significant authority and provides a high-level overview of the field, validating that my proposed solution is recognised by health systems internationally.

Research #4 (Rabiei and Almasi, BMC) is a systematic literature review published in a peer-reviewed academic journal. Its style is comprehensive and synthesising, bringing together findings from 54 separate studies to present a complete picture of dashboard requirements and challenges. This source provides a structured, evidence-based "blueprint" for designing hospital dashboards, offering both functional and non-functional requirements as well as categorised challenges and solutions.

Research #5 (Porta et al., Frontiers) is a study protocol for a large European research project. Its style is formal, detailed, and forward-looking, describing a planned study rather than completed results. This source is valuable because it shows that my proposed idea is being actively pursued by a major international consortium with significant funding (European Commission), demonstrating that the concept has real-world traction and legitimacy.

Research #6 (Wang et al., BMC) is a peer-reviewed research article that applies machine learning to predict ED wait times. Its style is technical and analytical, with a focus on algorithms, performance metrics, and model interpretability. This source introduces advanced predictive analytics into my inquiry, suggesting that my proposed outcome could evolve beyond simple real-time displays to include intelligent predictions based on factors like crowding and patient arrival mode.

Specific differences between sources:

The sources differ significantly in their scope and focus. Research #1 and #3 are broader, providing an overview of the problem and potential solutions at a systems level. Research #2 and #4 dive deep into technical implementation details, with #2 focusing on the granular challenges of data integration and #4 offering a structured framework for dashboard design. Research #5 and #6 are more forward-looking, with #5 describing a planned intervention and #6 demonstrating advanced predictive techniques.

The sources also differ in their geographical context. Research #1 is specific to New Zealand, making it directly relevant to my local inquiry. Research #2 is based in France, #3 in Canada, #4 is international, #5 spans six European countries, and #6 is based in the United States. This diversity strengthens my research by showing that ED overcrowding and the search for digital solutions are global issues, and that lessons can be drawn from multiple healthcare systems.

Finally, the sources differ in their level of evidence. Research #1 provides observational data from a single hospital. Research #2, #4, and #6 provide rigorous, peer-reviewed academic evidence. Research #3 provides a synthesis of existing evidence from a government agency. Research #5 describes a protocol for future evidence generation. Together, they create a rich, multi-layered foundation for my inquiry, combining local urgency with international expertise and rigorous academic validation.

Final Refined Inquiry Question:

I have refined my inquiry question, from:

"What are the key technical, ethical, and regulatory challenges involved in developing a real-time digital tool to display Emergency Department wait times in New Zealand, and how might these challenges be addressed to create a feasible and trustworthy public information system?"

To:

"What are the key technical, ethical, and regulatory challenges involved in developing a real-time digital dashboard to display Emergency Department wait times in New Zealand, and how might these challenges be addressed to create a feasible, trustworthy, and publicly accessible information system?"

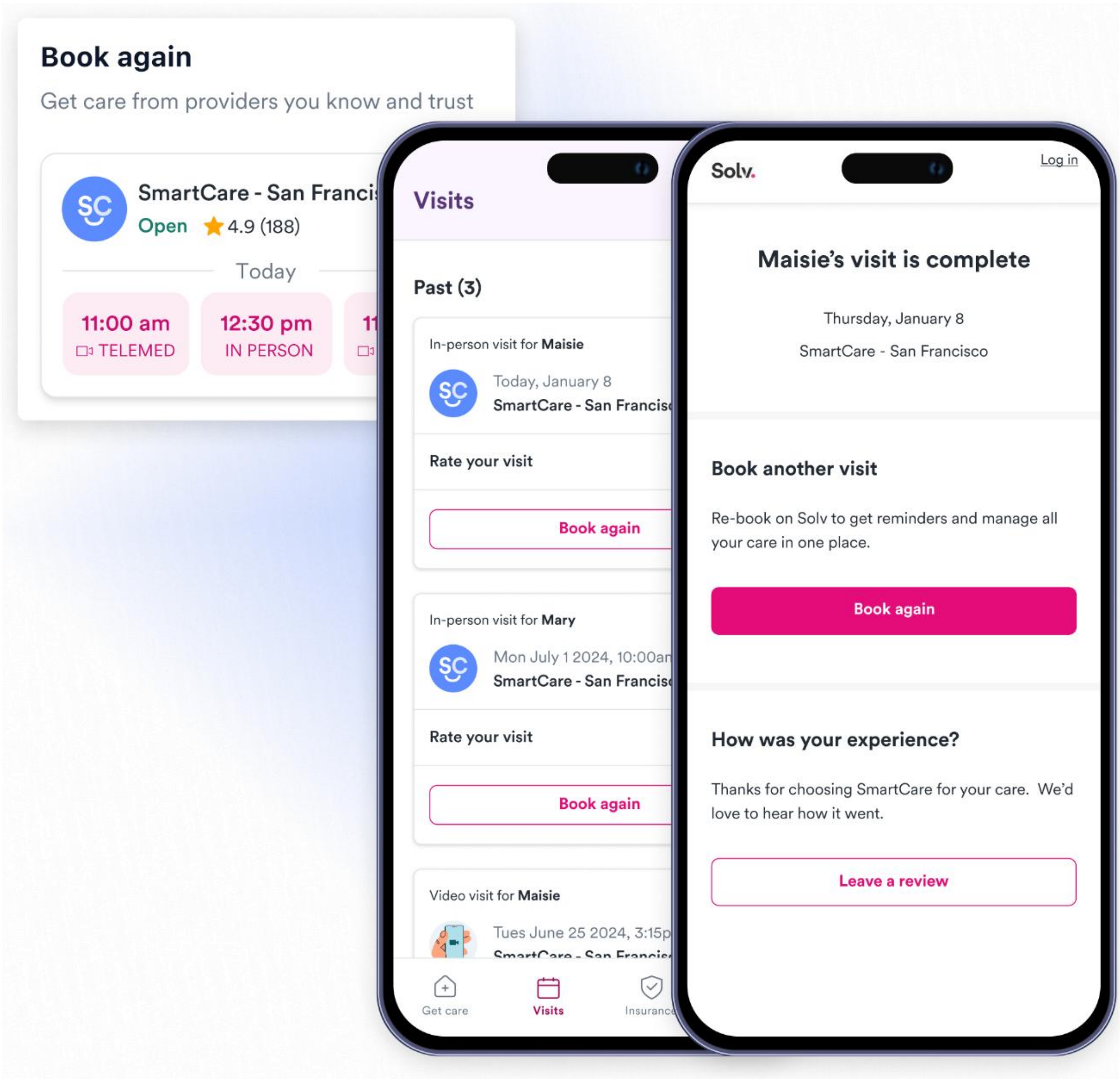
My initial inquiry question focused broadly on why people struggle to access ED wait time information and whether a digital tool could help. However, after analysing six diverse sources of evidence—ranging from local news reports to peer-reviewed academic studies and international government technology reviews—I have come to appreciate that the problem is far more complex than simply building an app. My research has revealed that while the need for such a tool is urgent and widely recognised, the path to creating one is filled with significant technical, ethical, and regulatory hurdles that must be carefully navigated.

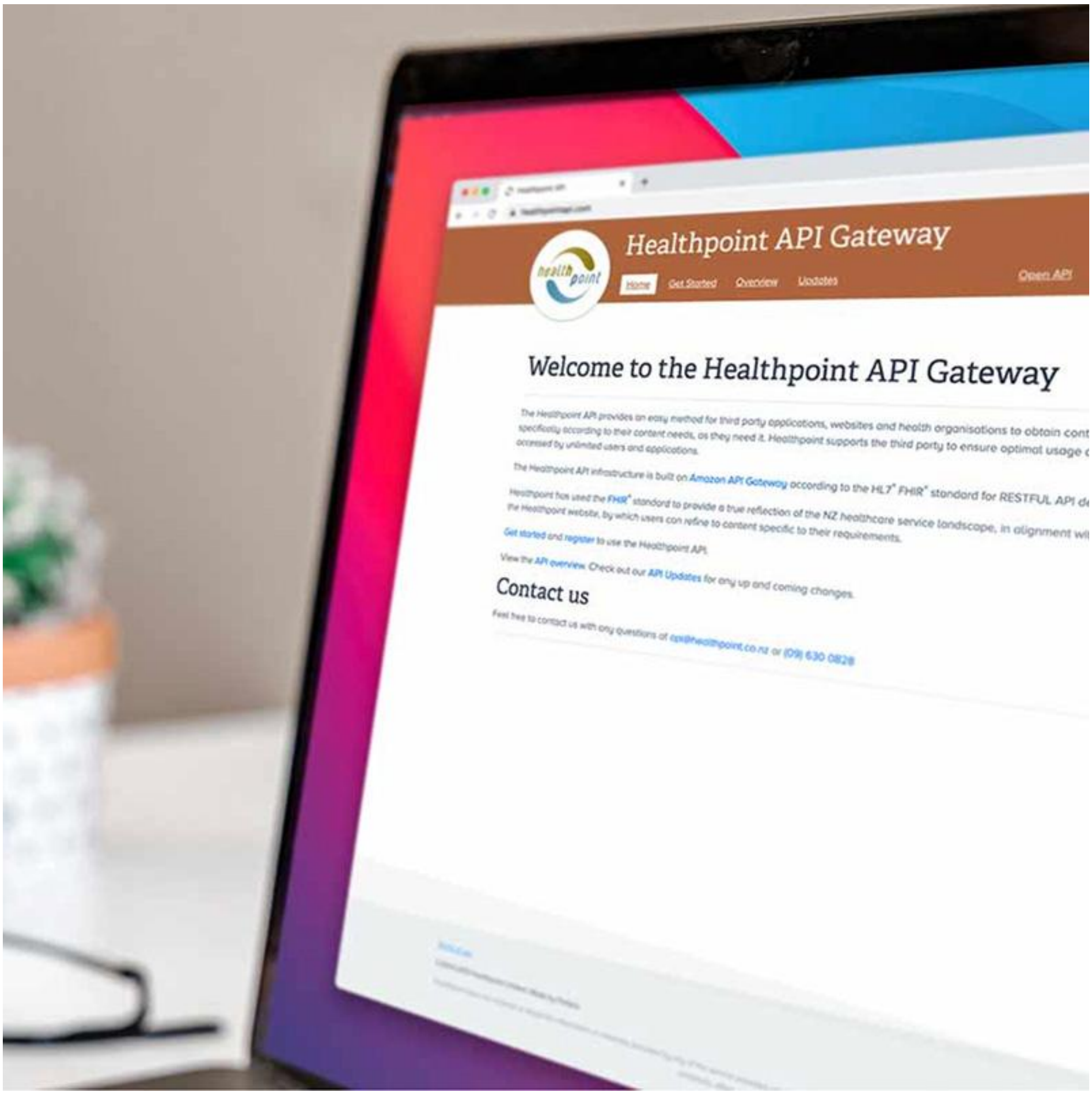
By refining my inquiry question to focus explicitly on the technical, ethical, and regulatory challenges, I am now able to move beyond simply proposing a digital solution. Instead, I can critically analyse the real-world barriers to implementation and propose a solution that is informed by evidence, grounded in an understanding of existing healthcare systems, and responsive to the legitimate concerns of stakeholders including clinicians, hospital administrators, privacy advocates, and the public.

This refined question will guide the final stages of my inquiry as I develop a proposal that acknowledges these complexities. Rather than presenting a simplistic "build an app" solution, I will propose a digital outcome that considers the need for data standardisation, stakeholder collaboration, user-centred design, robust privacy protections, and a phased implementation approach. By addressing these challenges head-on, my proposed outcome will be not only innovative but also feasible, trustworthy, and genuinely valuable to the New Zealand community.

Existing Solutions:

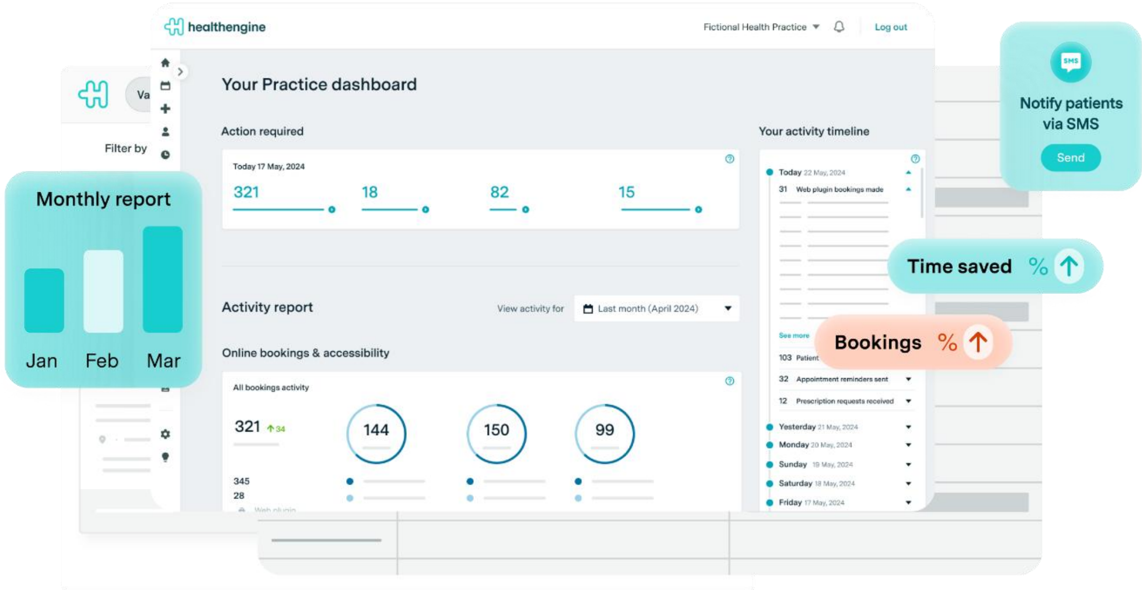
Existing Solution	Plus	Minus	Interesting
NHS 111 Online (UK)	Provides government-backed, publicly accessible estimated ED wait times for hospitals across England. Integrated into the wider NHS digital ecosystem, meaning users can access wait time data alongside symptom checking and GP finder tools in a single trusted platform. Free to use with no account required.	Wait time data is updated infrequently (sometimes only hourly), meaning it may not reflect the real-time situation inside the ED. The system is designed specifically for the NHS structure and cannot be directly replicated in New Zealand without equivalent government data-sharing agreements. Coverage is inconsistent — not all hospitals participate equally.	The NHS model demonstrates that a government agency can mandate data sharing from all hospitals and centralise it into one public platform. This is exactly the kind of institutional partnership that Health New Zealand (Te Whatu Ora) would need to facilitate for a New Zealand equivalent to succeed.
NHS 111 Online (UK) Images			

Solv Health (USA)	Allows users to view real-time wait times for urgent care clinics and some emergency departments in the United States. Users can book appointments directly through the app, and the interface is clean, mobile-friendly, and easy to navigate. Includes a map view showing nearby facilities and their current wait estimates, which is highly useful for decision-making.	Primarily built around the private US healthcare model, where clinics can opt in to list themselves. This means coverage is not universal — public hospitals may not participate. Not available in New Zealand. Wait time accuracy depends entirely on the clinic updating their own data, which creates reliability risks. The booking feature is less relevant in a publicly funded New Zealand context.	Solv shows that a commercial app can successfully combine geolocation, real-time data, and a simple map interface to help patients choose where to go. The colour-coded wait time display (green/yellow/red) is a design pattern worth adopting in my proposed outcome, as it communicates urgency without requiring the user to interpret numbers.
Solv Health (USA) Images			
Healthpoint NZ	Healthpoint.co.nz is the closest existing solution to my proposed outcome	Healthpoint does not display any real-time wait time data at all. It is a	Healthpoint proves that a government-backed NZ health information

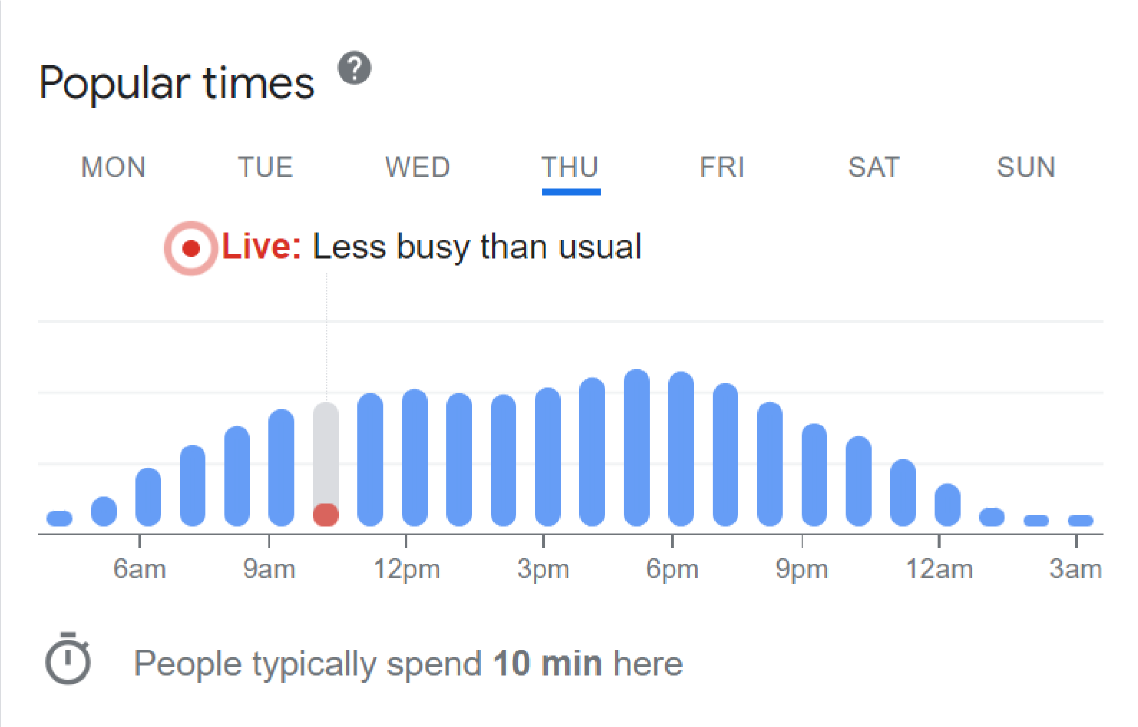
	within New Zealand. It is a government-supported healthcare directory that allows users to find local health services including emergency departments, after-hours clinics, and GPs. It provides basic information such as opening hours and contact details for all major NZ hospitals, and is free and publicly accessible.	static directory, not a live information system. There is no indication of how busy an ED is, how long a patient might wait, or how patient load compares across hospitals in the same region. The user interface is functional but dated and not well optimised for mobile use, which is how most people would access it in an emergency.	platform already exists and has public trust. This is significant because it means the infrastructure and institutional relationships are already partly in place. My proposed outcome could potentially be built as an extension or integration with Healthpoint, adding a real-time data layer on top of what already exists rather than starting from scratch.
Healthpoint NZ Images			
HealthEngine (Australia)	HealthEngine is Australia's largest health appointment booking platform. It	Wait time data on HealthEngine is self-reported by practices, not	The HealthEngine privacy controversy is a cautionary tale that

	displays estimated wait times for participating GP clinics and some after-hours services, and has a modern, mobile-first design that is highly rated for ease of use. It is Australia’s closest cultural and geographic equivalent to New Zealand’s healthcare context, making it a directly relevant reference point for my proposed outcome.	pulled from live hospital systems, meaning it can be inaccurate or outdated. Emergency departments are largely not covered — the platform focuses on bookable, non-urgent services. It has also faced criticism over data privacy practices after selling de-identified patient booking data to third parties without clear consent, which raises ethical concerns relevant to my own inquiry.	directly informs the ethical dimension of my proposed outcome. Any NZ equivalent must have explicit, transparent data governance policies from day one. The platform’s commercial success in Australia also shows there is strong public appetite for health information tools in similar healthcare contexts to New Zealand’s.

HealthEngine (Australia) Images



Google Maps — “Busy Times”	Google Maps already displays a “Live Busyness” indicator for many locations including some hospitals and clinics in New Zealand. It is universally available, requires no download, and is integrated into a tool most New Zealanders already use daily. The bar chart display of typical versus live busyness is instantly understandable and requires no health literacy to interpret.	The busyness data is crowd-sourced from users’ location data (via Android phones), not pulled from hospital clinical systems. This means it measures how many people’s phones are present in the building, not how many patients are in the waiting room or how long the actual clinical wait is. It cannot distinguish between staff, visitors, and patients, and is not reliable enough for urgent	The fact that people are already using Google Maps to check hospital busyness proves the public demand for this type of information in New Zealand. It also shows that a simple, visual indicator is more effective than detailed statistics for quick decision-making. My proposed outcome should aim to be at least as easy to interpret as Google’s busyness bar,

		healthcare decisions. It is also controlled entirely by Google with no transparency or accountability to the health sector.	but backed by real clinical data rather than passive location tracking.
Google Maps — "Busy Times" images			
Queensland Health ED Wait Times (Australia)	Queensland Health operates a publicly accessible online dashboard that displays real, near-time estimated wait times for emergency departments across Queensland. It is government funded, covers all major public hospitals in the state, and is updated every 15 minutes using actual triage data from hospital systems. This makes it the closest in the world to what my proposed outcome envisions for New Zealand.	The dashboard is web-based only and not well optimised for mobile use, which limits its usefulness in the moment a person actually needs it. The 15-minute update interval, while better than most alternatives, still means the data can be significantly out of date during fast-moving situations. It is not available as a dedicated app. The system required extensive government investment and legislative support to enable hospitals to share clinical data, which presents a significant barrier for replication in New Zealand	Queensland Health's dashboard is the strongest proof of concept for my proposed outcome. It shows that a public hospital system in a comparable country has already solved the data access and governance challenges that my research identified as the key barriers. It provides a real-world template for the data update frequency, interface design, and institutional partnerships my proposal would need. A key lesson is that government mandates — not voluntary participation —

		where similar mandates do not yet exist.	were essential to making the system work at scale.

Key Findings:

One of the clearest findings from this analysis is that no existing solution fully addresses the problem in the New Zealand context. The NHS and Queensland Health dashboards demonstrate that government-backed, data-integrated ED wait time tools are achievable and valued by the public, but both relied on legislative mandates and institutional investment that New Zealand does not yet have in place. Healthpoint NZ proves that a government-supported health information platform already exists here, but it is entirely static and lacks any real-time data layer. Solv and HealthEngine show that commercially-driven apps can make wait time information engaging and easy to use, but they are built around private healthcare booking models that do not translate to the New Zealand public health system. Google Maps' "Busy Times" feature is the most accessible option currently available to New Zealanders at no cost, but it is based on crowd-sourced location data rather than clinical information, making it unsuitable for genuine healthcare decision-making.

Together, these existing solutions confirm the gap my proposed outcome aims to fill: a mobile-first, publicly accessible, real-time ED information dashboard specifically designed for the New Zealand health system, built in partnership with Health New Zealand (Te Whatu Ora), and governed by transparent data privacy policies. The Queensland Health model in particular provides the most useful blueprint, while HealthEngine's privacy controversy highlights the ethical obligations any such tool must take seriously from the outset.

Discussion — Relevant Implications:

End User Consideration:

Explanation: The primary end users of the ER Population Tracker are members of the New Zealand public who are deciding where to seek urgent care. This includes people of all ages, digital literacy levels, and cultural backgrounds. Elderly users may have difficulty with complex interfaces. People in genuine medical distress will not have time to navigate confusing menus. Non-English speakers represent a significant proportion of users in cities like Auckland. The app must also consider secondary users — such as ambulance dispatchers, carers, and health officials — who may need different data views from the general public.

How it is relevant to my outcome: Understanding my end users shapes every design decision. Research #4 (Rabiei and Almasi) highlighted that successful dashboards require customisation and clear visualisation so different user groups can extract what they need quickly. My proposed outcome must therefore use a colour-coded, map-based interface (green/orange/red) that communicates wait severity at a glance, without requiring users to read detailed statistics. Font sizes, contrast ratios, and touch-target sizes should meet accessibility standards to serve elderly and visually impaired users. Multilingual support should be considered for a future iteration.

What research supports this: Research #4 (Rabiei and Almasi, BMC) found that effective hospital dashboards must include customisation for different user types and clear visual design. Research #5 (eCREAM) confirms that a separate "citizen-facing" dashboard with simpler information is necessary alongside a more detailed dashboard for clinicians and policymakers, specifically because the public needs different data from health professionals.

Functionality:

Explanation: The core function of the ER Population Tracker is to display real-time, or near-real-time, estimated wait times for all public emergency departments in a given region, ranked by proximity using geolocation. Beyond this, key functional requirements identified through research include: alerts for critically high wait times, tracking of patient flow over time, a map view with hospital markers colour-coded by urgency level, and the ability to filter by travel distance. The app would need to clearly define what "wait time" means (e.g., time from arrival to triage, or time to see a doctor) so users are not misled.

How it is relevant to my outcome: The functionality of my proposed outcome directly determines whether it solves the inquiry problem. If the wait time data is stale, mislabelled, or difficult to interpret, the app could cause more harm than good — for example, directing someone with a genuine emergency to a farther hospital based on incorrect data. This means the functionality requirements are not just technical but also ethical. Data must be labelled clearly, refreshed frequently, and accompanied by a disclaimer encouraging users to call 111 in life-threatening situations.

What research supports this: Research #4 (Rabiei and Almasi) identified tracking, alerting, and customisation as the three most important functional requirements for hospital dashboards. Research #1 (NZ Herald) noted that clinical staff themselves flagged the complexity of defining wait time, as patients in corridors may still be counted as "admitted." This directly informs my requirement to define and display wait time transparently.

Ethics and Privacy:

Explanation: Because the proposed outcome draws on data from hospital clinical systems, significant ethical and privacy obligations arise under New Zealand's Privacy Act 2020 and the Health Information Privacy Code. Even aggregated, de-identified wait time counts carry risk — for example, a small rural ED with very low patient numbers could inadvertently expose individual patient presence through a real-time count. There are also concerns about health equity: publicly displaying that a hospital is overwhelmed may discourage people with genuine emergencies from attending, creating a safety risk. Data governance policies must be established with Health New Zealand before any data sharing begins.

How it is relevant to my outcome: My proposed outcome must only display de-identified, aggregated data (patient counts and estimated waits, never individual identifiers). A data use agreement with Health New Zealand must clearly define the purpose of data collection as public information only, not clinical decision-making. The tool must include a visible disclaimer that it is not a medical advice service. HealthEngine's experience in Australia (see Existing Solutions) is a direct lesson: collecting health-adjacent data without a transparent privacy policy erodes public trust and can result in serious legal and reputational consequences.

What research supports this: Research #2 (Poirot et al.) specifically identified "secondary use" of patient data as a major ethical and regulatory concern, noting that hospital systems are not designed for external data sharing. Research #3 (CDA-AMC) called for explicit regulatory frameworks and privacy protections before ED information tools are deployed. Research #5 (eCREAM) addressed this by building data standardisation and a formal data governance protocol into the project from the outset.

Social Implications:

Explanation: An ER Population Tracker could have wide social benefits — reducing the frustration and uncertainty people experience when seeking urgent care, helping spread patient load more evenly across hospitals, and potentially reducing the number of people who leave without being seen due to long apparent wait times. However, there is also a risk of widening the digital divide: those without smartphones, reliable internet access, or digital literacy (disproportionately including elderly, low-income, and rural New Zealanders) may not be able to benefit from the tool, creating inequity in access to health information.

How it is relevant to my outcome: To mitigate the equity risk, my proposed outcome should be designed as both a mobile app and a web-based tool accessible on any browser, including on older devices. The tool must also be designed so that a carer or family member can easily check it on behalf of someone without a smartphone. Future iterations could explore integration with 111 dispatchers so that call handlers can use the tool to advise callers on which ED may have shorter waits.

Proposal:

My Proposal:

This proposal outlines the development of a real-time Emergency Department wait time dashboard for New Zealand, working title "WaitSmart NZ." The tool will allow any member of the New Zealand public to view current estimated wait times and patient load levels at nearby public emergency departments, displayed on an interactive colour-coded map, so they can make more informed decisions about where to seek urgent care.

Product Description:

WaitSmart NZ is a mobile-first, publicly accessible web application and smartphone app that displays near-real-time estimated emergency department wait times for all public hospitals participating in the scheme. Upon opening the app, users are shown a map of nearby EDs colour-coded by current status: green (under 1 hour), orange (1–3 hours), and red (over 3 hours or at critical capacity). Tapping any hospital marker reveals a detail view showing the current estimated wait, the number of patients currently in the waiting room, and a note defining what the wait time represents (time from arrival to triage). A prominent disclaimer at the top of every screen reads: "This tool is for information only. Call 111 if this is a life-threatening emergency." The app requires no login, no account, and no personal data from the user.

Target Audience:

The primary audience is the general New Zealand public facing a non-life-threatening urgent health concern — someone deciding whether to go to their nearest ED or an after-hours clinic, or choosing between multiple nearby hospitals. Secondary audiences include carers and family members assisting someone seeking care, and potentially 111 call handlers or ambulance dispatchers who could use the tool to give callers better guidance. The tool is designed to be usable by anyone with a smartphone or browser, with accessibility as a core design principle.

Requirements and Specifications:

Functional requirements include: real-time or near-real-time data display (target: updated every 10–15 minutes); geolocation to show the nearest EDs first; a colour-coded map view (green/orange/red); a hospital detail page showing current patient count and estimated wait with clear definitions; a 111 emergency call button prominently accessible from every screen; and a web version accessible on desktop and mobile browsers without an app download. Non-functional requirements include: page load under 3 seconds on a standard 4G connection; WCAG 2.1 AA accessibility compliance; data encryption in transit and at rest; and a clear privacy policy stating no user data is collected or stored.

Resources Required:

Developing WaitSmart NZ would require a cross-disciplinary team including frontend and backend software developers, a UX/UI designer with experience in accessibility and health information design, a data engineer to manage the API integration with hospital triage systems, and a health informatics specialist to oversee data standardisation and clinical accuracy. Institutionally, a formal data-sharing agreement and API access would need to be negotiated with Health New Zealand (Te Whatu Ora). The project would require cloud infrastructure for hosting and a data pipeline capable of ingesting and processing triage data from multiple hospitals on a 10–15 minute refresh cycle. A pilot phase focusing on a single region (e.g., Wellington or Auckland) would be the most realistic starting point before a national rollout.

Research Justification:

Research #1 (NZ Herald) confirms the urgent local need, with Wellington Hospital recording 575 code-red events and 3,200 patients leaving untreated in a single year. Research #3 (CDA-AMC) validates the solution type internationally, stating that "digital information tools provide real-time updates to patients about average wait times in the ED." Research #4 (Rabiei and Almasi) provides the evidence-based feature list for effective dashboards. Research #5 (eCREAM) demonstrates that a citizen-facing ED dashboard is not only feasible but is already being developed at scale in Europe. Research #6 (Wang et al.) shows that ED crowding is the single top predictor of prolonged wait times — the very metric my proposed tool would display — confirming that the data I am proposing to show is the most actionable information available. Finally, the Queensland Health dashboard (Existing Solutions) proves the concept works in a directly comparable public health system.

Future Possibilities:

In future iterations, WaitSmart NZ could incorporate predictive analytics — drawing on the machine learning approaches described in Research #6 (Wang et al.) — to forecast wait times over the next two to four hours, helping users plan ahead rather than just react to current conditions. Integration with after-hours clinic data (via Healthpoint) could allow the tool to suggest non-ED alternatives for lower-acuity conditions, reducing unnecessary ED presentations and further relieving system pressure. A secondary dashboard for Health New Zealand officials could display regional trends and crowding hotspots in real time, supporting policy and resourcing decisions. Push notifications could alert subscribed users when their nearest ED reaches critical capacity. Multilingual support would broaden access for non-English-speaking communities.

Risks and Risk Mitigation:

There are multiple risks for my proposed digital outcome that may limit its effectiveness or prevent it from being developed and deployed successfully. I need to identify and mitigate these risks so that my proposed outcome remains feasible, ethical, and genuinely valuable.

Data Access and Institutional Partnership Risk: The most significant risk is that Health New Zealand (Te Whatu Ora) may not agree to share live triage data, or that negotiating the required data-sharing agreement takes so long that the project stalls. Without this partnership, the core function of the app cannot exist. To mitigate this, the proposal could begin as a formal research collaboration with a university or public health institute, which may have existing relationships with Health NZ. A pilot approach — starting with a single willing hospital — would demonstrate value before seeking system-wide commitment. The Queensland Health precedent provides a compelling case study to present to decision-makers.

Data Accuracy and Reliability Risk: As Research #2 (Poirot et al.) showed in detail, hospital data is stored in multiple incompatible legacy systems that produce "messy," inconsistent data. If the wait time data displayed is inaccurate or stale, users could be misled into making a worse decision than if they had no information at all — for example, travelling to a hospital that was quiet when the app was last updated but is now overwhelmed. To mitigate this, the app should display the exact timestamp of the last data update prominently on every screen. A data validation layer in the backend should flag and quarantine anomalous readings before they are published. The 10–15 minute refresh target should be achievable in a pilot context, but the app should degrade gracefully (displaying a "data currently unavailable" message) rather than showing outdated information without a clear warning.

Misuse and Safety Risk: There is a risk that a person experiencing a genuine life-threatening emergency uses the app to choose a "shorter wait" hospital and delays calling 111 or going to the nearest ED. To mitigate this, every screen of the app must include a prominent emergency call button and a clear disclaimer. The onboarding screen (shown on first launch) must explain that the tool is for non-life-threatening urgent care decisions only, and that 111 should always be called for emergencies. User testing with diverse population groups during a pilot phase would help identify whether the current disclaimer design is sufficiently clear.

Privacy and Legal Risk: Even de-identified aggregate data about hospital patient counts is subject to the Privacy Act 2020 and the Health Information Privacy Code in New Zealand. Publishing real-time data that could be used to infer the presence of specific individuals (for example, in a small rural ED) creates legal exposure. To mitigate this, the data shared must always be aggregated to a minimum threshold (e.g., only displayed when there are five or more patients in the waiting room), and a Privacy Impact Assessment must be completed before the pilot launches. A formal legal review of the data-sharing agreement must be conducted with Health New Zealand's legal team.

Technical Development Risk: Building and maintaining a reliable real-time data pipeline between hospital systems and a public-facing app is technically complex and expensive. System outages, API failures, or changes to hospital software could cause the app to display incorrect or no data. To mitigate this, the app must have a robust monitoring and alerting system so that technical failures are detected and resolved quickly. A Service Level Agreement (SLA) with Health New Zealand should specify the expected data feed reliability. The app's backend should be hosted on a scalable cloud platform (e.g., AWS or Azure) with automatic failover. All source code should be version-controlled and thoroughly tested before each release.

Evaluation of the Proposed Outcome:

Strengths of the Proposed Outcome:

The strongest aspect of this proposal is that it is grounded in a real, urgent, and well-documented problem. Research #1 demonstrated with official OIA data that New Zealand's emergency departments are under severe strain, and that thousands of patients are leaving untreated each year. The proposed outcome directly addresses this by giving the public information they currently have no access to. The solution type is also internationally validated: Research #3 (CDA-AMC) and Research #5 (eCREAM) both confirm that digital ED information tools are recognised by government health agencies as viable and valuable. The existence of the Queensland Health dashboard proves the concept has already been implemented successfully in a comparable public health system, removing the risk that this is purely theoretical. The proposed outcome also builds on existing NZ infrastructure (Healthpoint) rather than starting entirely from scratch, which reduces both cost and institutional resistance. The design approach — mobile-first, colour-coded, no login required — maximises accessibility and minimises barriers to adoption.

Weaknesses of the Proposed Outcome:

The most significant weakness is the dependency on institutional cooperation that is entirely outside the control of the project team. Without a data-sharing agreement with Health New Zealand, the app cannot function at all. This is not a technical problem that can be solved by better engineering — it is a regulatory and political challenge that requires sustained advocacy and relationship-building. Research #2 (Poirot et al.) showed that even willing hospitals face enormous internal technical challenges in making their data accessible, meaning the timeline for a fully functional system could be years rather than months. A second weakness is the equity gap: people without smartphones or reliable internet access — who may include some of the most vulnerable patients — would not be able to benefit from the tool. A third weakness is the risk of misuse: if the app is used by someone in a genuine emergency to delay calling 111, the consequences could be severe. Finally, there is an inherent tension between data freshness and accuracy — the more frequently the data updates, the higher the infrastructure cost and the greater the risk of displaying a momentary anomaly as if it were the true state of the ED.

Further Opportunities and Areas for Improvement:

The most important area for follow-up is conducting user testing with a diverse group of New Zealanders during the pilot phase — including elderly users, people with disabilities, and users from non-English-speaking backgrounds — to verify that the interface is genuinely accessible and the emergency disclaimer is sufficiently prominent. The data definitions used by the tool (particularly what "wait time" means) should be validated with clinical staff to ensure they are accurate and not misleading. Future versions could incorporate after-hours clinic data from Healthpoint, creating a more complete picture of urgent care options and actively redirecting lower-acuity patients away from EDs. As machine learning models for wait time prediction mature (as discussed in Research #6), integrating a predictive “forecast” feature would significantly increase the tool’s usefulness. Measuring the tool’s actual impact on ED attendance patterns — for example, whether it leads to a measurable redistribution of patients across hospitals — would be important for evaluating its real-world effectiveness and justifying continued investment.

Does the Proposed Outcome Answer the Inquiry Focus?

Yes. My refined inquiry question asked: “What are the key technical, ethical, and regulatory challenges involved in developing a real-time digital dashboard to display Emergency Department wait times in New Zealand, and how might these challenges be addressed to create a feasible, trustworthy, and publicly accessible information system?” My proposed outcome directly addresses this by identifying the three key challenge categories — data access and standardisation (technical), privacy and misuse (ethical), and institutional partnership (regulatory) — and proposing specific, evidence-based mitigations for each. The outcome is designed to be publicly accessible (no login, web and app versions), trustworthy (clear data timestamps, transparent definitions, Privacy Act compliance), and feasible (piloted in one region before national rollout, built on existing NZ health infrastructure). The research conducted throughout this inquiry — particularly Research #2, #3, and #5 — has directly shaped these design decisions, meaning the proposal is evidence-led rather than speculative.

Summary of the Inquiry Process:

This inquiry began with a broad personal observation: when my family has needed to visit a hospital emergency department, we have never known in advance how long the wait would be or whether a different hospital nearby might have been a better choice. From this everyday frustration, I developed my initial inquiry question about why New Zealanders cannot access real-time emergency department wait time information, and whether a digital tool could solve this problem.

The research phase of this inquiry transformed my understanding of the problem. I initially assumed the main challenge would be building the app itself. Research #2 (Poirot et al.) was a turning point — it showed that even willing hospitals struggle to make their data accessible because it is stored in multiple incompatible legacy systems. This shifted my question away from “how do we build it?” toward “what are the real barriers, and how can they be addressed?” This is what led to my refined inquiry question, which now explicitly focuses on the technical, ethical, and regulatory challenges. The research also broadened my perspective significantly: I had not initially considered the privacy implications of displaying patient count data, the health equity risks of a smartphone-only tool, or the importance of defining what “wait time” actually means clinically.

The stakeholder perspectives I gathered confirmed the genuine public demand for this type of tool. The findings showed that most people currently rely on guesswork when deciding which ED to attend, and that the majority would use a real-time wait time app if one existed. Some stakeholders also raised concerns about data accuracy and the risk of misuse, which directly informed the risk mitigation strategies in my proposal. The analysis of existing solutions was equally important: seeing that Queensland Health has already built and deployed a functional ED wait time dashboard for the public proved that my proposed outcome is not just a good idea but an achievable one.

The planning process also taught me important lessons. Using Trello to manage my tasks and milestones kept me on track across a long project and made it easier to see where I was falling behind and adjust. The critical inquiry cycle — moving from brainstorming to research to analysis to proposal — proved to be a genuinely useful structure. Each stage built meaningfully on the last: the brainstorm informed the initial question, the research refined it, the analysis of existing solutions sharpened the proposal, and the risk assessment grounded it in reality.

If I were to do this inquiry again, I would seek out a direct interview with a Health New Zealand data manager or a hospital IT professional early in the process, as this would have given me more accurate insight into the real-world data access challenges much sooner. I would also conduct user testing with real participants rather than relying solely on a form, as in-person testing reveals usability issues that survey responses do not capture. Overall, however, I am confident that my proposed outcome — WaitSmart NZ — is a meaningful, evidence-based, and feasible response to a genuine gap in New Zealand’s public health information infrastructure.

Acknowledgements

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